

Quantum Field Theory

Course notes

Francisco Gómez Serrano

Units: $\hbar = c = 1$. Part I uses the metric $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$;
Part II follows Srednicki and uses $g^{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ (see its introduction).

Contents

I	Canonical Quantisation	4
1	Introduction and Classical Field Theory	4
1.1	Review: the action	4
1.2	Hamiltonian formalism	5
1.3	Symmetries and conserved “charges”	5
2	Quantisation of the Free Scalar Field	6
2.1	Internal symmetries (continuation)	6
2.2	The free real scalar field	7
2.3	Quantisation of the harmonic oscillator	7
2.4	Quantisation of the free scalar field	7
3	The Complex Scalar Field	8
3.1	Heisenberg picture: causality and propagators	9
4	Propagators and Interacting Scalar Fields	10
4.1	The Feynman propagator	10
4.2	Pictures	11
4.3	Dyson formula	11
4.4	Scattering processes	12
4.5	Wick’s theorem	12
5	Interactions and Feynman Diagrams	13
5.1	Review	13
5.2	Interaction terms and dimensional analysis	13
5.3	Scattering in ϕ^4 theory	13

6	Interacting Scalar Field and Feynman Diagrams	14
6.1	Second order	15
6.2	Feynman diagrams and rules	15
6.3	Why do we discard disconnected diagrams and vacuum bubbles?	16
7	Interacting Scalar Fields and Fermions	16
7.1	Scalar Yukawa theory	16
7.2	Mandelstam variables (for $2 \rightarrow 2$)	17
7.3	Relation to observables	18
8	Fermions	18
8.1	Groups and representations	18
8.2	The Lorentz group $SO(3,1)$	19
8.3	$SU(2)$ and spinors	19
8.4	Lorentz-invariant Lagrangian for Weyl and Dirac spinors	20
9	Fermions II	20
9.1	Noether currents	20
9.2	Quantisation of the Dirac field	21
9.3	Internal symmetries	22
9.4	Feynman propagator for fermions	22
9.5	Yukawa theory with fermions	22
9.6	Gauge fields	23
9.7	Discrete symmetries for fermions	25
10	Quantum Electrodynamics	25
10.1	Some processes	26
II	Path Integrals and Renormalisation	27
11	Poincaré Symmetry and the Free Field Revisited	27
11.1	The Poincaré algebra	27
11.2	Canonical quantisation in the mostly-plus metric	28
12	The LSZ Reduction Formula	29
13	Path Integrals in Quantum Mechanics	30
13.1	Time evolution and the Dyson formula	30
13.2	Derivation of the path integral	31
13.3	The forced harmonic oscillator	31

14 Path Integrals for Scalar Fields and Feynman Diagrams	33
14.1 From mechanics to fields	33
14.2 Renormalised Lagrangian and counterterms	33
14.3 Feynman diagrams from $Z[J]$	34
14.4 Scattering amplitude at tree level	35
15 Dimensional Analysis and Renormalisability	36
16 The Exact Propagator and the Self-Energy	36
16.1 Källén–Lehmann representation	36
16.2 Self-energy	37
16.3 Dimensional regularisation	38
17 Vertex Correction and Loop Integrals	39
18 Power Counting and the Quantum Action	40
18.1 Superficial degree of divergence	40
18.2 ϕ^4 theory	41
18.3 Skeleton expansion and the quantum action	41
19 Infrared Divergences and Observables	43
20 The Renormalisation Group	44
20.1 Physical mass and the $\overline{\text{MS}}$ scheme	44
20.2 Running coupling	45
20.3 Renormalisation group equations from the counterterms	46

Part I

Canonical Quantisation

1 Introduction and Classical Field Theory

The goal is to describe quantum and relativistic phenomena for systems in which creation and annihilation of particles happen (the number of particles is not conserved). **Fields** are the fundamental quantity; **particles** are excited states of fields.

Let us confine a particle in a box of length $\Delta x = L$. Then

$$\Delta p \gtrsim \frac{\hbar}{L} \implies \Delta E \gtrsim \frac{\hbar c}{L}.$$

If $\Delta E \gtrsim 2mc^2$, the vacuum starts creating/destroying (anti)particles, which happens for

$$L \lesssim \lambda_c = \frac{\hbar}{mc},$$

the **Compton wavelength**. Two approaches to quantisation: canonical quantisation and the path integral.

Remarks. Fields Φ can be scalar (real ϕ or complex φ), vector (A_μ) or spinor (ψ). We use $\hbar = c = 1$ and Minkowski space with $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$.

1.1 Review: the action

The action is

$$S = \int dt L = \int dt d^3x \mathcal{L}, \quad \mathcal{L} = \mathcal{L}(\Phi, \partial_\mu \Phi),$$

with L the Lagrangian and $\mathcal{L}$ the **Lagrangian density**. (Higher derivative terms end up in instabilities of the system.) $\mathcal{L}$ must yield second-order equations of motion, be Poincaré invariant (no explicit dependence on the coordinates), and respect the symmetries of the specific problem. Varying,

$$\begin{aligned} \delta S &= \int_{\mathcal{M}} d^4x \left[\frac{\partial \mathcal{L}}{\partial \Phi} \delta \Phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \partial_\mu (\delta \Phi) \right], \\ \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \partial_\mu \delta \Phi &= \underbrace{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \delta \Phi \right)}_{\text{boundary term}} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \right) \delta \Phi. \end{aligned}$$

The boundary term can be neglected as long as $\delta \Phi|_{\partial \mathcal{M}} = 0$, so

$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \Phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \right) \right] \delta \Phi,$$

and $\delta S = 0$ gives the **Euler–Lagrange equations of motion**.

Free massive scalar field.

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 = \underbrace{\frac{1}{2}\dot{\phi}^2}_{\sim T} - \underbrace{\frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2}_{\sim V},$$

whose equation of motion is the **Klein–Gordon equation**

$$\partial_\mu\partial^\mu\phi + m^2\phi = 0, \quad \square := \partial_\mu\partial^\mu \text{ (d'Alembertian).}$$

Complex scalar field (equivalent to two real scalar fields, $\varphi = (\phi_1 + i\phi_2)/\sqrt{2}$):

$$\mathcal{L} = (\partial_\mu\varphi)(\partial^\mu\varphi)^* - m^2\varphi\varphi^*.$$

Varying with respect to φ^* and φ :

$$\square\varphi + m^2\varphi = 0, \quad \square\varphi^* + m^2\varphi^* = 0.$$

Electromagnetic vector potential. For a vector field $A_\mu = (\phi, \vec{A})$, the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ contains $\vec{E}$ and $\vec{B}$. With

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J_\mu A^\mu, \quad J^\mu = (\rho, \vec{J}),$$

the equations of motion are $\partial_\mu F^{\mu\nu} = J^\nu$.

1.2 Hamiltonian formalism

The **conjugate momentum** and the **Hamiltonian density** are

$$\pi(x) := \frac{\partial\mathcal{L}}{\partial\dot{\Phi}}, \quad \mathcal{H}(\Phi, \pi) = \pi\dot{\Phi} - \mathcal{L},$$

and Hamilton's equations of motion read

$$\dot{\Phi} = \frac{\partial\mathcal{H}}{\partial\pi}, \quad \dot{\pi} = -\frac{\partial\mathcal{H}}{\partial\Phi} + \partial_i\left(\frac{\partial\mathcal{H}}{\partial(\partial_i\Phi)}\right).$$

1.3 Symmetries and conserved “charges”

Statement (Noether's theorem). Every continuous symmetry of the Lagrangian gives rise to a conserved current, namely a vector J^μ such that $\partial_\mu J^\mu = 0$ on-shell (imposing the equations of motion).

For an infinitesimal transformation $\delta\Phi = X(\Phi)$ that is a symmetry, $\delta\mathcal{L} = \partial_\mu F^\mu$, and

$$J^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)}\delta\Phi - F^\mu, \quad Q = \int d^3x J^0 \quad \text{with} \quad \frac{dQ}{dt} = 0.$$

Types of symmetries:

- Poincaré: Lorentz transformations and translations.
- Internal: symmetries of the field (e.g. in the weak and electromagnetic interactions).

Examples: translations give the energy–momentum tensor; Lorentz transformations give the angular momentum (which differs for spinors).

Energy-momentum tensor. Under $x^\mu \rightarrow x'^\mu = x^\mu + \varepsilon^\mu$ (active transformation, $\phi'(x') = \phi(x)$),

$$\delta\phi = \phi'(x) - \phi(x) = \phi(x - \varepsilon) - \phi(x) = -\varepsilon^\nu \partial_\nu \phi, \quad \delta\mathcal{L} = -\varepsilon^\nu \partial_\nu \mathcal{L},$$

so

$$J^\mu = \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \delta_\nu^\mu \mathcal{L} \right] (-\varepsilon^\nu), \quad T^\mu{}_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \delta_\nu^\mu \mathcal{L},$$

the **energy-momentum tensor**. It can be symmetrised ($T_{\mu\nu} \propto \delta S / \delta g^{\mu\nu}$). The conserved charges are

$$E = \int d^3x T^{00} = \int d^3x (\pi \dot{\phi} - \mathcal{L}), \quad P^i = \int d^3x T^{0i} = - \int d^3x \pi \partial_i \phi.$$

Lorentz transformations. $x'^\mu = \Lambda^\mu{}_\nu x^\nu$ with $\Lambda^\top \eta \Lambda = \eta$ (ds^2 is invariant). Infinitesimally $\Lambda^\mu{}_\nu = \delta^\mu_\nu + \omega^\mu{}_\nu$, and $\Lambda^\top \eta \Lambda = \eta$ requires $\omega^{\mu\nu} = -\omega^{\nu\mu}$. Then

$$\begin{aligned} \delta\phi &= \phi'(x) - \phi(x) = \phi(\Lambda^{-1}x) - \phi(x) = \phi(x - \omega x) - \phi(x) = -\omega^\mu{}_\nu x^\nu \partial_\mu \phi, \\ \delta\mathcal{L} &= -\omega^\mu{}_\nu x^\nu \partial_\mu \mathcal{L} = -\partial_\mu (\omega^\mu{}_\nu x^\nu \mathcal{L}), \end{aligned}$$

which gives the conserved currents

$$(\mathcal{J}^\mu)^{\rho\sigma} = x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho}.$$

For $\rho = 0, \sigma = i$ (boosts):

$$Q^{0i} = \int d^3x (x^0 T^{0i} - x^i T^{00}), \quad \frac{dQ^{0i}}{dt} = P^i + t \underbrace{\frac{dP^i}{dt}}_0 - \frac{d}{dt} \int d^3x x^i T^{00} = 0,$$

and for the rotations $Q^{ij} = \int d^3x (x^i T^{0j} - x^j T^{0i})$.

2 Quantisation of the Free Scalar Field

2.1 Internal symmetries (continuation)

The symmetry acts on the field. For the complex scalar field $\mathcal{L} = (\partial_\mu \varphi)(\partial^\mu \varphi)^* - m^2 \varphi \varphi^*$:

$$\varphi \rightarrow e^{i\alpha} \varphi, \quad \varphi^* \rightarrow e^{-i\alpha} \varphi^* \quad \implies \quad \delta\varphi = i\alpha\varphi, \quad \delta\varphi^* = -i\alpha\varphi^*.$$

Since $\delta\mathcal{L} = 0$ ($F^\mu = 0$),

$$J^\mu = i((\partial^\mu \varphi^*)\varphi - (\partial^\mu \varphi)\varphi^*), \quad Q = \int d^3x J^0 = i \int d^3x (\dot{\varphi}^* \varphi - \dot{\varphi} \varphi^*).$$

Passive and active transformations. ϕ is a scalar iff $\phi'(x') = \phi(x)$, and

$$\delta_P \phi = \phi(x') - \phi(x) \quad (\text{passive}), \quad \delta_A \phi = \phi'(x) - \phi(x) \quad (\text{active}).$$

2.2 The free real scalar field

A free scalar field can be seen as an infinite collection of harmonic oscillators. One way to see it: a lattice model of coupled oscillators becomes the Klein–Gordon field in the continuum limit $N \rightarrow \infty$. Another way: start from the unique Poincaré-invariant action

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2 \implies (\Box + m^2)\phi = (\partial_t^2 - \nabla^2 + m^2)\phi = 0,$$

where ∇^2 couples the field at different points. In Fourier space,

$$\phi(\vec{x}, t) = \int \frac{d^3 p}{(2\pi)^3} \phi_{\vec{p}}(t) e^{i\vec{p} \cdot \vec{x}} \implies \partial_t^2 \phi_{\vec{p}}(t) + \underbrace{(\vec{p}^2 + m^2)}_{\omega_{\vec{p}}^2} \phi_{\vec{p}}(t) = 0 :$$

the field is an infinite collection of decoupled harmonic oscillators. Now one just needs to quantise $\phi_{\vec{p}}(t)$ (or $\phi(\vec{x}, t)$).

2.3 Quantisation of the harmonic oscillator

$$L = \frac{\dot{q}^2}{2} - \frac{\omega^2}{2} q^2 \implies \ddot{q} + \omega^2 q = 0, \quad p = \frac{\partial L}{\partial \dot{q}} = \dot{q}, \quad H = \frac{p^2}{2} + \frac{\omega^2 q^2}{2},$$

with p and q now operators satisfying $[q, p] = i\hbar \mathbb{K}$. The “ladder” (creation and annihilation) operators

$$a = \sqrt{\frac{\omega}{2\hbar}} q + \frac{i}{\sqrt{2\omega\hbar}} p, \quad a^\dagger = \sqrt{\frac{\omega}{2\hbar}} q - \frac{i}{\sqrt{2\omega\hbar}} p, \quad [a, a^\dagger] = \mathbb{K},$$

give

$$H = \frac{\hbar\omega}{2}(aa^\dagger + a^\dagger a) = \hbar\omega \left(a^\dagger a + \frac{1}{2}\mathbb{K} \right),$$

where $a^\dagger$ to the left and a to the right is **normal ordering**. From $[H, a^\dagger] = \hbar\omega a^\dagger$ and $[H, a] = -\hbar\omega a$,

$$H(a^\dagger |n\rangle) = (E_n + \hbar\omega)a^\dagger |n\rangle, \quad H(a |n\rangle) = (E_n - \hbar\omega)a |n\rangle,$$

$$|n\rangle = \frac{1}{\sqrt{n!}}(a^\dagger)^n |0\rangle, \quad \langle n|n\rangle = 1.$$

2.4 Quantisation of the free scalar field

With $\pi_{\vec{p}}(t) = \partial \mathcal{L} / \partial \dot{\phi}_{\vec{p}}(t)$, the Hamiltonian density is

$$\mathcal{H} = \frac{\pi^2}{2} + \frac{(\nabla \phi)^2}{2} + \frac{m^2 \phi^2}{2}.$$

In the Schrödinger picture (for now) we promote $\phi_{\vec{p}}$ and $\pi_{\vec{p}}$ to (time-independent) operators,

$$\phi_{\vec{p}} = \frac{1}{\sqrt{2\omega_{\vec{p}}}}(a_{\vec{p}} + a_{-\vec{p}}^\dagger), \quad \pi_{\vec{p}} = -i\sqrt{\frac{\omega_{\vec{p}}}{2}}(a_{\vec{p}} - a_{-\vec{p}}^\dagger),$$

where the combination $a_{\vec{p}} + a_{-\vec{p}}^\dagger$ ensures that $\phi(\vec{x})$ is real ($\phi_{\vec{p}}^* = \phi_{-\vec{p}}$). The commutation relations

$$[a_{\vec{p}}, a_{\vec{q}}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}), \quad [a_{\vec{p}}, a_{\vec{q}}] = 0 = [a_{\vec{p}}^\dagger, a_{\vec{q}}^\dagger]$$

(equivalently $[\phi_{\vec{p}}, \pi_{\vec{q}}] = i(2\pi)^3 \delta^{(3)}(\vec{p} + \vec{q})$) encode the spin statistics of the field. The **mode expansion** is

$$\begin{aligned}\phi(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} (a_{\vec{p}} + a_{-\vec{p}}^\dagger) e^{i\vec{p}\cdot\vec{x}} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} (a_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + a_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}}), \\ \pi(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} (-i) \sqrt{\frac{\omega_{\vec{p}}}{2}} (a_{\vec{p}} - a_{-\vec{p}}^\dagger) e^{i\vec{p}\cdot\vec{x}}.\end{aligned}$$

The Hamiltonian

$$H = \frac{1}{2} \int d^3x (\pi^2 + (\nabla\phi)^2 + m^2\phi^2) = \int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} \left(a_{\vec{p}}^\dagger a_{\vec{p}} + \frac{1}{2} (2\pi)^3 \delta^{(3)}(0) \right)$$

contains an infinite constant: $\delta^{(3)}(0) \rightarrow \infty$, and moreover $\omega_{\vec{p}} \sim p$ so $\int d^3p \omega_{\vec{p}} \rightarrow \infty$. The vacuum $|0\rangle$, with $a_{\vec{k}}|0\rangle = 0$, is the ground state, with energy

$$E_0 = \langle 0|H|0\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{(2\pi)^3 \delta^{(3)}(0)}{2} \omega_{\vec{p}}.$$

With $\delta^{(3)}(0) = \int d^3x e^{-i\vec{p}\cdot\vec{x}}|_{\vec{p}=0} = V$, the vacuum energy density is

$$\varepsilon_0 = \frac{E_0}{V} = \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} = \frac{1}{4\pi^2} \int_0^\infty dp p^2 \sqrt{p^2 + m^2} \rightarrow \infty.$$

The energy density still diverges. Partial solution: an energy cut-off, since we only care about energy differences. (In general relativity one cares about the total energy, not only differences.)

We therefore use **normal ordering**:

$$:H: = \int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} a_{\vec{p}}^\dagger a_{\vec{p}} \implies \langle 0|:H:|0\rangle = 0.$$

Particles. From $[H, a_{\vec{p}}] = -\omega_{\vec{p}} a_{\vec{p}}$ and $[H, a_{\vec{p}}^\dagger] = \omega_{\vec{p}} a_{\vec{p}}^\dagger$, the states

$$|\vec{p}\rangle := \sqrt{2\omega_{\vec{p}}} a_{\vec{p}}^\dagger |0\rangle, \quad \langle \vec{p}_1 | \vec{p}_2 \rangle = 2\omega_{\vec{p}} (2\pi)^3 \delta^{(3)}(\vec{p}_1 - \vec{p}_2),$$

are momentum eigenstates, and

$$H |\vec{p}\rangle = \sqrt{2\omega_{\vec{p}}} \int \frac{d^3q}{(2\pi)^3} \omega_{\vec{q}} a_{\vec{q}}^\dagger ((2\pi)^3 \delta^{(3)}(\vec{q} - \vec{p}) + a_{\vec{p}}^\dagger a_{\vec{q}}) |0\rangle = \omega_{\vec{p}} |\vec{p}\rangle :$$

$|\vec{p}\rangle$ is an eigenstate of the Hamiltonian with eigenvalue $\omega_{\vec{p}} = E_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}$. Multi-particle states $|\vec{p}_1, \dots, \vec{p}_n\rangle \propto \prod_i (a_{\vec{p}_i}^\dagger)^{n_i} |0\rangle$ belong to the **Fock space** $\mathcal{F} = \bigoplus_n \mathcal{H}_n$, a generalisation of the Hilbert space.

3 The Complex Scalar Field

$$\mathcal{L} = \partial_\mu \varphi \partial^\mu \varphi^* - m^2 \varphi \varphi^*, \quad \varphi = \frac{\phi_1 + i\phi_2}{\sqrt{2}}, \quad \pi = \dot{\varphi}^*, \quad \pi^* = \dot{\varphi}.$$

Promoting φ and π to operators,

$$\begin{aligned}\varphi(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} (b_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + c_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}}), \\ \pi(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} i \sqrt{\frac{\omega_{\vec{p}}}{2}} (b_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}} - c_{\vec{p}} e^{i\vec{p}\cdot\vec{x}}),\end{aligned}$$

with $b_{\vec{p}} = \frac{1}{\sqrt{2}}(a_{1,\vec{p}} + ia_{2,\vec{p}})$ and $c_{\vec{p}} = \frac{1}{\sqrt{2}}(a_{1,\vec{p}} - ia_{2,\vec{p}})$.

Commutation relations:

$$\begin{aligned} [\varphi(\vec{x}), \pi(\vec{y})] &= i\delta^{(3)}(\vec{x} - \vec{y}), & [\varphi^\dagger(\vec{x}), \pi^\dagger(\vec{y})] &= i\delta^{(3)}(\vec{x} - \vec{y}), \\ [b_{\vec{p}}, b_{\vec{q}}^\dagger] &= (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}), & [c_{\vec{p}}, c_{\vec{q}}^\dagger] &= (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}). \end{aligned}$$

The conserved charge due to the symmetry $\varphi \rightarrow e^{i\alpha}\varphi$ is

$$Q = i \int d^3x (\dot{\varphi}^* \varphi - \varphi^* \dot{\varphi}) = i \int d^3x (\pi\varphi - \varphi^* \pi^*),$$

which after normal ordering becomes

$$:Q: = \int \frac{d^3p}{(2\pi)^3} (c_{\vec{p}}^\dagger c_{\vec{p}} - b_{\vec{p}}^\dagger b_{\vec{p}}) = N_c - N_b,$$

and likewise (following D. Tong's QFT lecture notes)

$$:H: = \int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} (c_{\vec{p}}^\dagger c_{\vec{p}} + b_{\vec{p}}^\dagger b_{\vec{p}}), \quad P^i = \int \frac{d^3p}{(2\pi)^3} p^i (b_{\vec{p}}^\dagger b_{\vec{p}} + c_{\vec{p}}^\dagger c_{\vec{p}}).$$

3.1 Heisenberg picture: causality and propagators

From the Schrödinger to the Heisenberg picture,

$$O_H(t) = e^{iHt} O_S e^{-iHt}, \quad \frac{dO_H}{dt} = i[H, O_H].$$

With $[H, a_{\vec{p}}] = -\omega_{\vec{p}} a_{\vec{p}}$ and $[H, a_{\vec{p}}^\dagger] = \omega_{\vec{p}} a_{\vec{p}}^\dagger$,

$$a_{\vec{p}}(t) = e^{-i\omega_{\vec{p}}t} a_{\vec{p}}, \quad a_{\vec{p}}^\dagger(t) = e^{i\omega_{\vec{p}}t} a_{\vec{p}}^\dagger,$$

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} (a_{\vec{p}} e^{-ip \cdot x} + a_{\vec{p}}^\dagger e^{ip \cdot x}) \Big|_{p^0 = E_{\vec{p}} = \omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}},$$

where $p^0 = E_{\vec{p}}$ is the **mass-shell condition**. Note that $e^{-ip \cdot x} = e^{-ip^0 t} e^{i\vec{p} \cdot \vec{x}}$ (the $e^{+ip^0 t}$ terms are the negative-frequency states). The equal-time commutation relation is

$$[\phi(\vec{x}, t), \pi(\vec{y}, t)] = i\delta^{(3)}(\vec{x} - \vec{y}),$$

and

$$\phi(x) = e^{i(Ht - P_i x^i)} \phi(0) e^{-i(Ht - P_i x^i)}, \quad P_\mu = (H, P_j).$$

Causality. In order to have a *causal* theory,

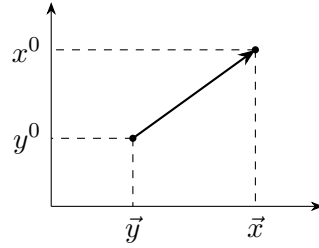
$$[\phi(x), \phi(y)] = 0 \quad \forall (x - y)^2 < 0 \text{ (spacelike)}.$$

Define

$$\Delta(x - y) := \langle 0 | [\phi(x), \phi(y)] | 0 \rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} (e^{-ip \cdot (x-y)} - e^{ip \cdot (x-y)}).$$

For spacelike separations one can choose a frame with $x^0 = y^0$, where the two terms cancel ($\vec{p} \rightarrow -\vec{p}$), so $\Delta = 0$; for timelike separations $(t, \vec{0})$ it behaves as $e^{-imt} - e^{imt} \neq 0$ (stationary phase approximation). Causality is obeyed thanks to the existence of particle-antiparticle pairs.

Propagation amplitude. Prepare a particle at the spacetime point y : what is the amplitude of finding it at x ?



$$\begin{aligned} D(x-y) &:= \langle 0 | \phi(x) \phi(y) | 0 \rangle = \int \frac{d^3 p}{(2\pi)^3} \frac{d^3 q}{(2\pi)^3} \frac{1}{\sqrt{4E_{\vec{p}} E_{\vec{q}}}} \langle 0 | a_{\vec{p}} a_{\vec{q}}^\dagger | 0 \rangle e^{-ip \cdot x} e^{iq \cdot y} \\ &= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} e^{-ip \cdot (x-y)}. \end{aligned}$$

For timelike separations $D \sim e^{-imt}$; for spacelike separations $D \sim e^{-m|\vec{x}-\vec{y}|} \neq 0$: there is leakage outside the light cone. However, $\Delta(x-y) = D(x-y) - D(y-x)$ vanishes there.

4 Propagators and Interacting Scalar Fields

4.1 The Feynman propagator

$$\Delta_F(x-y) := \langle 0 | T \{ \phi(x) \phi(y) \} | 0 \rangle,$$

where the **time-ordering operator** moves to the left all operators evaluated at later times:

$$T \{ \phi(x) \phi(y) \} = \begin{cases} \phi(x) \phi(y), & x^0 > y^0, \\ \phi(y) \phi(x), & x^0 < y^0. \end{cases}$$

Hence

$$\begin{aligned} \Delta_F(x-y) &= \theta(x^0 - y^0) D(x-y) + \theta(y^0 - x^0) D(y-x) \\ &= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} \left[\theta(x^0 - y^0) e^{-ip \cdot (x-y)} + \theta(y^0 - x^0) e^{ip \cdot (x-y)} \right], \end{aligned}$$

a particle propagating from y to x or from x to y (the first bit of Feynman diagrams). Equivalently, integrating for the first time over p^0 ,

$$\Delta_F(x-y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x-y)}, \quad \tilde{\Delta}_F(p) = \frac{i}{p^2 - m^2 + i\epsilon},$$

the **Feynman propagator in momentum space**.

Proof. $p^2 - m^2 = (p^0)^2 - E_{\vec{p}}^2 = (p^0 + E_{\vec{p}})(p^0 - E_{\vec{p}})$, so the integrand has two poles, displaced off the real axis by the $i\epsilon$ prescription.

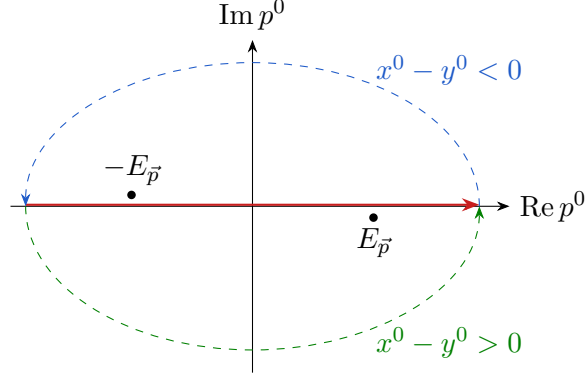


Figure 4.1: Poles of the Feynman propagator and the contours used for $x^0 > y^0$ (closing below) and $x^0 < y^0$ (closing above).

By the residue theorem, for $x^0 - y^0 > 0$ (closing clockwise below),

$$\int \frac{dp^0}{2\pi} \frac{i}{p^2 - m^2} e^{-ip^0(x^0 - y^0)} = \frac{1}{2E_{\vec{p}}} e^{-iE_{\vec{p}}(x^0 - y^0)},$$

and for $x^0 - y^0 < 0$ (closing above) one picks up the pole at $-E_{\vec{p}}$, giving $\frac{1}{2E_{\vec{p}}} e^{iE_{\vec{p}}(x^0 - y^0)}$. Combining with the spatial integral (and $\vec{p} \rightarrow -\vec{p}$ in the second term) reproduces the expression above. ■

4.2 Pictures

- Schrödinger picture: operators are time independent, states are time dependent, $|\psi\rangle = |\psi(t)\rangle$.
- Heisenberg picture: operators are time dependent, $O = O(t)$; states are not.
- Interaction picture: $H = H_0 + H_1$, with H_0 the free theory and H_1 a small perturbation; both operators and states depend on time.

In the interaction picture,

$$O_I(t) = e^{iH_0 t} O e^{-iH_0 t} \implies \frac{dO_I}{dt} = i[H_0, O_I], \quad |\psi_I(t)\rangle := e^{iH_0 t} |\psi(t)\rangle.$$

Expectation values agree, $\langle \psi(t) | O | \psi(t) \rangle = \langle \psi_I(t) | O_I(t) | \psi_I(t) \rangle$, and

$$\begin{aligned} i\partial_t |\psi_I(t)\rangle &= e^{iH_0 t} (-H_0 + H) e^{-iH_0 t} |\psi_I(t)\rangle \implies \\ i\partial_t |\psi_I(t)\rangle &= H_I(t) |\psi_I(t)\rangle, \quad H_I(t) := e^{iH_0 t} H_1 e^{-iH_0 t}. \end{aligned}$$

4.3 Dyson formula

With $|\psi_I(t)\rangle = U(t, t_0) |\psi_I(t_0)\rangle$ (H_I at different times need not commute),

$$i \frac{d}{dt} U(t, t_0) = H_I(t) U(t, t_0).$$

Solving perturbatively:

- order 0: $i \frac{dU^{(0)}}{dt} = 0 \Rightarrow U^{(0)} = \mathbb{I}$;
- order 1: $i \frac{dU^{(1)}}{dt} = H_I(t) U^{(0)} \Rightarrow U^{(1)} = -i \int_{t_0}^t dt' H_I(t')$;
- order 2: $i \frac{dU^{(2)}}{dt} = H_I(t) U^{(1)} \Rightarrow U^{(2)} = - \int_{t_0}^t dt \int_{t_0}^{\tilde{t}} dt' H_I(\tilde{t}) H_I(t')$,

a multiple integral with nested boundaries. Relabelling the integration region (the triangle $t_0 < t_2 < t_1 < t$),

$$U^{(2)} = - \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H_I(t_1) H_I(t_2) = -\frac{1}{2} \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 T\{H_I(t_1) H_I(t_2)\},$$

and in general

$$U^{(n)} = \frac{1}{n!} T \left\{ \left(-i \int_{t_0}^t dt' H_I(t') \right)^n \right\} \implies U(t, t_0) = T \exp \left(-i \int_{t_0}^t dt' H_I(t') \right),$$

the **Dyson formula**.

4.4 Scattering processes

We want scattering cross sections, matrices and amplitudes. $\langle f|S|i \rangle$ is the amplitude for the state $|i \rangle$ ($t \rightarrow -\infty$) to evolve into the final state $|f \rangle$ ($t \rightarrow +\infty$) via the unitary **S-matrix**, which describes the interaction:

$$\langle f|S|i \rangle := \lim_{\substack{t_0 \rightarrow -\infty \\ t \rightarrow +\infty}} \langle f|U(t, t_0)|i \rangle, \quad S = T \exp \left(-i \int_{-\infty}^{\infty} dt H_I(t) \right).$$

Writing $S = \mathbb{1} + i\mathcal{T}$,

$$\langle f|i\mathcal{T}|i \rangle = (2\pi)^4 \delta^{(4)}(p_f - p_i) i\mathcal{A}.$$

4.5 Wick's theorem

With $N\{\dots\} := :(\dots):$ and the **contraction** $\overline{\phi_1 \phi_2} := \Delta_F(x_1 - x_2)$, $\phi_i := \phi(x_i)$,

$$T\{\phi_1 \phi_2 \dots \phi_n\} = N\{\phi_1 \phi_2 \dots \phi_n + \text{all possible contractions}\}.$$

For example,

$$T\{\phi_1 \phi_2 \phi_3 \phi_4\} = N\{\phi_1 \phi_2 \phi_3 \phi_4 + \overline{\phi_1 \phi_2} \phi_3 \phi_4 + \overline{\phi_1 \phi_3} \phi_2 \phi_4 + \dots + \overline{\phi_1 \phi_2} \overline{\phi_3 \phi_4} + \overline{\phi_1 \phi_3} \overline{\phi_2 \phi_4} + \overline{\phi_1 \phi_4} \overline{\phi_2 \phi_3}\}.$$

Since the vacuum expectation value of any normal-ordered product vanishes,

$$\langle 0|T\{\phi_1 \dots \phi_n\}|0 \rangle = \text{all possible full contractions}.$$

For $n = 4$,

$$\begin{aligned} \langle 0|T\{\phi_1 \phi_2 \phi_3 \phi_4\}|0 \rangle &= \Delta_F(x_1 - x_2) \Delta_F(x_3 - x_4) \\ &\quad + \Delta_F(x_1 - x_3) \Delta_F(x_2 - x_4) + \Delta_F(x_1 - x_4) \Delta_F(x_2 - x_3), \end{aligned}$$

represented by the three diagrams pairing the four points.

Proof for two fields. Split $\phi(x) = \phi^+(x) + \phi^-(x)$ with

$$\phi^+(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} a_{\vec{p}} e^{-ip \cdot x}, \quad \phi^-(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} a_{\vec{p}}^\dagger e^{ip \cdot x}.$$

Then

$$\begin{aligned} N\{\phi_1 \phi_2\} &= \phi_1^+ \phi_2^+ + \phi_1^- \phi_2^- + \phi_1^- \phi_2^+ + \phi_2^- \phi_1^+, \\ T\{\phi_1 \phi_2\} &= \phi_1^+ \phi_2^+ + \phi_1^- \phi_2^- + \theta(t_1 - t_2) [\phi_1^+ \phi_2^- + \phi_1^- \phi_2^+] + \theta(t_2 - t_1) [\phi_2^+ \phi_1^- + \phi_2^- \phi_1^+], \end{aligned}$$

so

$$T\{\phi_1 \phi_2\} - N\{\phi_1 \phi_2\} = \theta(t_1 - t_2) [\phi_1^+, \phi_2^-] + \theta(t_2 - t_1) [\phi_2^+, \phi_1^-] = \Delta_F(x_1 - x_2),$$

a c-number. ■

5 Interactions and Feynman Diagrams

5.1 Review

$$H_I = e^{iH_0 t} H_1 e^{-iH_0 t}, \quad |\psi_I(t)\rangle = U_I(t, t_0) |\psi_I(t_0)\rangle,$$

$$i \frac{dU_I}{dt} = H_I U_I \implies U_I(t, t_0) = T \exp \left(-i \int_{t_0}^t dt' H_I(t') \right),$$

$$\langle f|S|i\rangle = \lim_{\substack{t_+ \rightarrow +\infty \\ t_- \rightarrow -\infty}} \langle f|U(t_+, t_-)|i\rangle, \quad S = \lim_{\substack{t_+ \rightarrow +\infty \\ t_- \rightarrow -\infty}} T \exp \left(-i \int_{t_-}^{t_+} dt H_I(t) \right).$$

To evaluate $\langle f|T\{\phi(x_1) \cdots \phi(x_n)\}|i\rangle$, Wick's theorem will be useful.

5.2 Interaction terms and dimensional analysis

Interactions can be added, in particular

$$\Delta\mathcal{L} = \mathcal{L}_{\text{int}} = - \sum_{n \geq 3} \frac{\lambda_n}{n!} \phi^n.$$

For a given energy only a finite number of these terms are relevant. With $\hbar = c = 1$ the action is dimensionless, $[S] = 0$ in mass dimensions ($[m] = 1$), so $[\mathcal{L}] = 4$; since $[\partial_\mu] = 1$, $[\phi] = 1$ and

$$[\lambda_n] = 4 - n.$$

It is convenient to use dimensionless couplings, $\tilde{\lambda}_n = \lambda_n/E^{4-n}$, where E is the relevant energy scale.

- $n = 3$: the ϕ^3 term is negligible at large energies but important at low energies: **relevant** term.
- $n = 4$: ϕ^4 matters at all energies: **marginal** term.
- $n > 4$: ϕ^n is important only at high energy scales: **irrelevant** term.

ϕ^4 theory.

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4!} \phi^4.$$

Yukawa theory (a toy-model approximation of QED):

$$\mathcal{L}_Y = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \partial_\mu \psi^* \partial^\mu \psi - M^2 \psi^* \psi - g \psi^* \psi \phi.$$

5.3 Scattering in ϕ^4 theory

With $H_1 = \frac{\lambda}{4!} \phi^4$ and

$$|i\rangle = \sqrt{4E_{\vec{p}_1} E_{\vec{p}_2}} a_{\vec{p}_1}^\dagger a_{\vec{p}_2}^\dagger |0\rangle = |\vec{p}_1 \vec{p}_2\rangle, \quad |f\rangle = \sqrt{4E_{\vec{p}_3} E_{\vec{p}_4}} a_{\vec{p}_3}^\dagger a_{\vec{p}_4}^\dagger |0\rangle = |\vec{p}_3 \vec{p}_4\rangle,$$

the scattering amplitude is

$$\begin{aligned} \langle f|S|i\rangle &= \langle f| \lim T \left\{ \exp \left(-i \int dt d^3x \mathcal{H}_1 \right) \right\} |i\rangle \\ &= \langle \vec{p}_3 \vec{p}_4 | \vec{p}_1 \vec{p}_2 \rangle - \frac{i\lambda}{4!} \int d^4x \langle \vec{p}_3 \vec{p}_4 | T \{ \phi^4(x) \} | \vec{p}_1 \vec{p}_2 \rangle \\ &\quad - \frac{1}{2} \left(\frac{\lambda}{4!} \right)^2 \int d^4x d^4y \langle \vec{p}_3 \vec{p}_4 | T \{ \phi^4(x) \phi^4(y) \} | \vec{p}_1 \vec{p}_2 \rangle + \cdots \end{aligned}$$

Zeroth order.

$$\begin{aligned}\langle \vec{p}_3 \vec{p}_4 | \vec{p}_1 \vec{p}_2 \rangle &= 4\sqrt{E_1 E_2 E_3 E_4} \langle 0 | a_{\vec{p}_3} a_{\vec{p}_4} a_{\vec{p}_1}^\dagger a_{\vec{p}_2}^\dagger | 0 \rangle \\ &= 4E_{\vec{p}_1} E_{\vec{p}_2} (2\pi)^6 [\delta^{(3)}(\vec{p}_3 - \vec{p}_1) \delta^{(3)}(\vec{p}_4 - \vec{p}_2) \\ &\quad + \delta^{(3)}(\vec{p}_3 - \vec{p}_2) \delta^{(3)}(\vec{p}_4 - \vec{p}_1)] : \end{aligned}$$

the two particles pass by without interacting.

First order. By Wick's theorem,

$$\langle \vec{p}_3 \vec{p}_4 | T\{\phi_1 \phi_2 \phi_3 \phi_4\} | \vec{p}_1 \vec{p}_2 \rangle = \langle \vec{p}_3 \vec{p}_4 | N\{\phi_1 \phi_2 \phi_3 \phi_4 + \text{all possible contractions}\} | \vec{p}_1 \vec{p}_2 \rangle,$$

where uncontracted fields act on the external states:

$$\phi^+(x) |\vec{p}\rangle = \int \frac{d^3 k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{k}}}} a_{\vec{k}} e^{-ik \cdot x} \sqrt{2E_{\vec{p}}} a_{\vec{p}}^\dagger | 0 \rangle = e^{-ip \cdot x} | 0 \rangle.$$

For $\phi^4(x)$ there are three kinds of terms (all at the same point x):

(A) two internal contractions: a **vacuum bubble** times the free propagation,

$$S_1^A = -\frac{i\lambda}{4!} \cdot 3 \langle \vec{p}_3 \vec{p}_4 | \vec{p}_1 \vec{p}_2 \rangle \int d^4 x \Delta_F^2(x - x);$$

(B) one internal contraction (a loop on a line): these are **disconnected diagrams**, where a particle goes from the initial to the final state along the same line,

$$S_1^B = -\frac{i\lambda}{4!} \langle \vec{p}_4 | \vec{p}_2 \rangle \int d^4 x \Delta_F(x - x) e^{i(p_3 - p_1) \cdot x} + 3 \text{ other terms};$$

(C) all four fields contracted with the external states, with a combinatorial factor $4!$:

$$S_1^C = 4! \left(-\frac{i\lambda}{4!}\right) \int d^4 x e^{i(p_3 + p_4 - p_1 - p_2) \cdot x} = -i\lambda (2\pi)^4 \delta^{(4)}(p_3 + p_4 - p_1 - p_2).$$

Hence the **tree-level scattering amplitude** is $i\mathcal{A} = -i\lambda$. The vacuum bubbles are eliminated by the normalisation with respect to the vacuum $|\Omega\rangle$ of the interacting theory (see below), and disconnected diagrams do not contribute to scattering. Writing $S = \mathbb{K} + i\mathcal{T}$,

$$\langle f | i\mathcal{T} | i \rangle = (2\pi)^4 \delta^{(4)}(p_f - p_i) i\mathcal{A}.$$

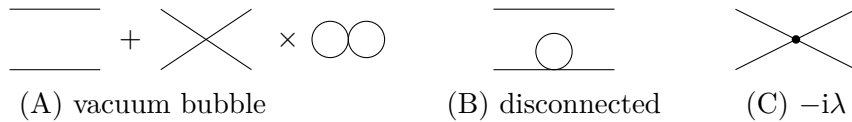


Figure 5.1: First-order contributions to $2 \rightarrow 2$ scattering in ϕ^4 theory.

6 Interacting Scalar Field and Feynman Diagrams

(Peskin–Schroeder 4.2, 4.4, 4.6.)

6.1 Second order

Since we are only interested in full contractions with the external fields,

$$S_2 = \frac{1}{2} \left(-\frac{i\lambda}{4!} \right)^2 \int d^4x d^4y \langle \vec{p}_3 \vec{p}_4 | \phi_x \phi_x \phi_x \phi_x \phi_y \phi_y \phi_y \phi_y | \vec{p}_1 \vec{p}_2 \rangle$$

with the external momenta p_1, p_2 contracted at y , p_3, p_4 at x , and two propagators between x and y . Counting: $12 \cdot 12$ ways to attach the external lines, 2 ways to join the remaining fields, and 2 for $x \leftrightarrow y$, i.e. $576 = (4!)^2$. Hence

$$S_2 = \frac{(-i\lambda)^2}{2} \int d^4x d^4y \Delta_F(x-y) \Delta_F(x-y) e^{i(p_3+p_4) \cdot x} e^{-i(p_1+p_2) \cdot y}.$$

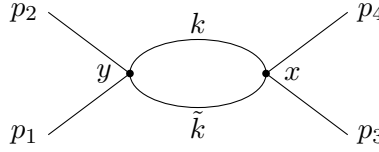


Figure 6.1: One-loop (s -channel) contribution in ϕ^4 theory.

In momentum space,

$$S_2 = \frac{(-i\lambda)^2}{2} \int d^4x d^4y e^{-i(p_1+p_2) \cdot y} e^{i(p_3+p_4) \cdot x} \int \frac{d^4k}{(2\pi)^4} \frac{ie^{-ik \cdot (x-y)}}{k^2 - m^2 + i\epsilon} \int \frac{d^4\tilde{k}}{(2\pi)^4} \frac{ie^{-i\tilde{k} \cdot (x-y)}}{\tilde{k}^2 - m^2 + i\epsilon},$$

which gives

$$S_2 = (2\pi)^4 \delta^{(4)}(p_3 + p_4 - p_1 - p_2) \frac{(-i\lambda)^2}{2} \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} \frac{i}{(p_1 + p_2 - k)^2 - m^2 + i\epsilon},$$

$$i\mathcal{A}_2 = \frac{(-i\lambda)^2}{2} \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} \frac{i}{(p_1 + p_2 - k)^2 - m^2 + i\epsilon}.$$

For large k , $\mathcal{A}_2 \sim \int^\Lambda d^4k k^{-4} \sim \ln \Lambda$: a logarithmic divergence.

6.2 Feynman diagrams and rules

Steps (position space or momentum space):

1. Draw all (relevant) diagrams: draw an external line for each particle in $|i\rangle$ and $|f\rangle$, and join the external lines using the vertices given by H_I .
2. Assign an amplitude:
 - each vertex contributes $-i\lambda$ (for this theory), together with an integration over all spacetime;
 - each internal line ($x \rightarrow y$) contributes $\Delta_F(x-y)$ (in momentum space $\tilde{\Delta}_F(k)$, with an integration over unconstrained internal momenta);
 - each external line contributes $e^{-ip \cdot x}$ (incoming) or $e^{ip \cdot x}$ (outgoing) in position space, and 1 in momentum space;
 - divide by the symmetry factor;
 - impose momentum conservation at each vertex.

6.3 Why do we discard disconnected diagrams and vacuum bubbles?

(Peskin–Schroeder 4.2, 4.6.) The amplitude

$$\langle f | (S - \mathcal{K}) | i \rangle = \lim_{t_{\pm} \rightarrow \pm\infty} \left(\langle f | T \left\{ e^{-i \int d^4x \mathcal{H}_I} \right\} | i \rangle \right)_0$$

is computed with free states and the free vacuum, keeping only external lines fully connected. With $H = H_0 + H_I$, the interaction-picture field is the free field, $\phi_I(x) = e^{iH_0 t} \phi(\vec{x}) e^{-iH_0 t}$, related to the Heisenberg field by $\phi(x) = U^{-1}(t, 0) \phi_I(x) U(t, 0)$, and $\phi_I(x) |\vec{p}\rangle = e^{-ip \cdot x} |0\rangle$ involves the free vacuum, $H_0 |0\rangle = 0$. It should be the vacuum $|\Omega\rangle$ of the interacting theory, $H |\Omega\rangle = E_0 |\Omega\rangle$: the two are different, $|0\rangle \neq |\Omega\rangle$, but they overlap, $\langle \Omega | 0 \rangle \neq 0$. Evolving adiabatically,

$$|\Omega\rangle = \lim_{T \rightarrow \infty(1-i\epsilon)} (e^{-iE_0 T} \langle \Omega | 0 \rangle)^{-1} e^{-iHT} |0\rangle, \quad |\vec{p}_1 \vec{p}_2\rangle = \lim_{T \rightarrow \infty(1-i\epsilon)} e^{-iHT} |\vec{p}_1 \vec{p}_2\rangle_0,$$

and

$$\langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle = \lim_{T \rightarrow \infty(1-i\epsilon)} \frac{\langle 0 | T \{ \phi_I(x) \phi_I(y) \exp(-i \int_{-T}^T dt H_I(t)) \} | 0 \rangle}{\langle 0 | T \{ \exp(-i \int_{-T}^T dt H_I(t)) \} | 0 \rangle},$$

where the denominator, made of vacuum bubble diagrams (the “particle sea”), cancels the vacuum bubbles of the numerator.

7 Interacting Scalar Fields and Fermions

7.1 Scalar Yukawa theory

$$\mathcal{L} = \underbrace{\frac{1}{2}(\partial_\mu \phi)^2 - \frac{m^2}{2}\phi^2}_{\text{“photon-like”}} + \underbrace{\partial_\mu \psi^* \partial^\mu \psi - M^2 \psi^* \psi}_{\text{“electron-like”}} - \underbrace{g \psi^* \psi \phi}_{\text{vertex}}.$$

In this case the coupling has dimensions of energy, $[g] = 1$; in the fermion version (the useful one) g is dimensionless. In diagrams, ϕ is drawn as a dashed line and ψ as a solid line, with arrows indicating the flow of charge.

“Electron” scattering $\psi\psi \rightarrow \psi\psi$. At order g^2 , with $|i\rangle = \sqrt{4E_{\vec{p}_1} E_{\vec{p}_2}} b_{\vec{p}_1}^\dagger b_{\vec{p}_2}^\dagger |0\rangle = |\vec{p}_1 \vec{p}_2\rangle$ and $|f\rangle = |\vec{p}_3 \vec{p}_4\rangle$,

$$\langle f | S | i \rangle = \frac{(-ig)^2}{2} \int d^4x d^4y \langle \vec{p}_3 \vec{p}_4 | T \{ (\psi^* \psi \phi)_x (\psi^* \psi \phi)_y \} | \vec{p}_1 \vec{p}_2 \rangle.$$

Contracting $\phi_x \phi_y$ and the ψ ’s with the external states (plus $x \leftrightarrow y$, which cancels the $\frac{1}{2}$), with $\psi(x) |\vec{p}\rangle = e^{-ip \cdot x} |0\rangle$, $\langle \vec{p} | \psi^*(x) = \langle 0 | e^{ip \cdot x}$,

$$\begin{aligned} \langle f | S | i \rangle &= (-ig)^2 \int d^4x d^4y e^{i(p_3 - p_1) \cdot x} e^{i(p_4 - p_2) \cdot y} \int \frac{d^4q}{(2\pi)^4} \frac{ie^{-iq \cdot (x-y)}}{q^2 - m^2} \\ &= (-ig)^2 \int \frac{d^4q}{(2\pi)^4} \frac{i}{q^2 - m^2} (2\pi)^4 \delta^{(4)}(p_3 - p_1 - q) (2\pi)^4 \delta^{(4)}(p_4 - p_2 + q) \\ &= i(-ig)^2 \frac{1}{(p_1 - p_3)^2 - m^2} (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4), \end{aligned}$$

$$\mathcal{A}_t = (-ig)^2 \frac{1}{(p_1 - p_3)^2 - m^2} \quad (t\text{-channel}), \quad \mathcal{A}_u = (-ig)^2 \frac{1}{(p_1 - p_4)^2 - m^2} \quad (u\text{-channel}),$$

the second coming from the other contraction ($p_3 \leftrightarrow p_4$).

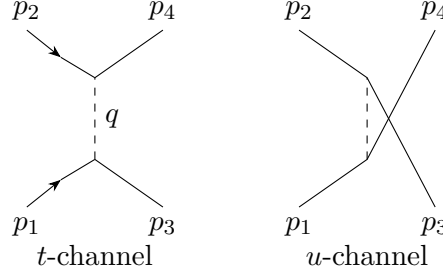


Figure 7.1: Order- g^2 diagrams for $\psi\psi \rightarrow \psi\psi$ in scalar Yukawa theory.

7.2 Mandelstam variables (for $2 \rightarrow 2$)

$$s := (p_1 + p_2)^2, \quad t := (p_1 - p_3)^2, \quad u := (p_1 - p_4)^2,$$

where s is the centre-of-mass energy squared and t, u measure the momentum exchange (and include the scattering angle). They satisfy

$$\sum_i m_i^2 = s + t + u.$$

Why is $i\epsilon$ not included in these propagators? Before, it was needed because of the on-shell condition; now the particle in between is virtual and does not necessarily satisfy the on-shell condition. Be careful with the s -channel, where $s = m^2$ is possible: add $i\epsilon$.

Feynman rules for scalar Yukawa theory.

- each vertex: $-ig$;
- each external line: 1;
- internal lines: $\frac{i}{k^2 - m^2 + i\epsilon}$ (dashed), $\frac{i}{k^2 - M^2 + i\epsilon}$ (solid);
- momentum conservation at each vertex;
- integrate over all unconstrained internal momenta.

Some diagrams.

- (a) Scattering of a ψ particle off a ϕ particle ($\psi\phi \rightarrow \psi\phi$), exchanging a ψ (two diagrams):

$$\mathcal{A} = (-ig)^2 \left(\frac{1}{t - M^2} + \frac{1}{u - M^2} \right),$$

with t and u defined by the momentum q flowing through the internal line in each diagram.

- (b) $\psi\bar{\psi} \rightarrow \psi\bar{\psi}$ (“electron-positron” scattering): t -channel and s -channel,

$$\mathcal{A} = (-ig)^2 \left(\frac{1}{t - m^2} + \frac{1}{s - m^2 + i\epsilon} \right).$$

The s -channel shows resonances, which is how new particles are found.

- (c) $\phi\phi \rightarrow \phi\phi$ (“photon” scattering): at order g^4 the simplest diagram involves a loop,

$$\mathcal{A} = (-ig)^4 \int \frac{d^4k}{(2\pi)^4} \frac{1}{D_0 D_1 D_{12} D_3},$$

with $D_0 = k^2 - M^2 + i\epsilon$, $D_1 = (k - p_1)^2 - M^2 + i\epsilon$, $D_{12} = (k - p_1 - p_2)^2 - M^2 + i\epsilon$ and $D_3 = (k - p_3)^2 - M^2 + i\epsilon$.

7.3 Relation to observables

Decay rates, cross sections and correlations. The Lorentz-invariant phase-space element for N final particles is

$$d\Pi := (2\pi)^4 \delta^{(4)}(p_f - p_i) \prod_{n=1}^N \frac{d^3 p_n}{(2\pi)^3} \frac{1}{2E_{\vec{p}_n}}.$$

Decay rate of a particle of mass M (in its rest frame):

$$\Gamma_0 = \frac{1}{2M} \sum_f \int |\mathcal{A}|^2 d\Pi,$$

where $\int |\mathcal{A}|^2 d\Pi$ is Lorentz invariant but the prefactor must be corrected in a moving frame ($E = \gamma M$, $\gamma = 1/\sqrt{1-v^2}$): $\Gamma = \Gamma_0/\gamma$. The half-life $\tau_0 = \Gamma_0^{-1}$ becomes $\tau = \gamma\tau_0$.

Cross section. The rate of scattering is $R = \sigma \mathcal{L}$, where the luminosity $\mathcal{L}$ is the flux of particles per unit area per unit time and σ the scattering cross section. For $2 \rightarrow N$,

$$\sigma = \frac{1}{4I} \sum_f \int |\mathcal{A}|^2 d\Pi, \quad I = \sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2} \quad (\text{flux factor}).$$

I is frame dependent, and so is σ ; it is usually calculated in the centre-of-mass frame. For $2 \rightarrow 2$ in the centre-of-mass frame, only the solid-angle integration is left:

$$d\Pi = \frac{1}{(2\pi)^2} \frac{|\vec{p}'|}{4\sqrt{s}} d\Omega, \quad s = (p_1 + p_2)^2 = E_{\text{cm}}^2,$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\vec{p}'|}{|\vec{p}|} |\mathcal{A}|^2 \quad (\vec{p} \text{ incoming, } \vec{p}' \text{ outgoing}), \quad \left(\frac{d\sigma}{d\Omega} \right)_{\text{cm}}^{\text{elastic}} = \frac{|\mathcal{A}|^2}{64\pi^2 s}.$$

Peaks in the cross section signal particles.

8 Fermions

8.1 Groups and representations

- $SU(2)$: the group of 2×2 unitary matrices with determinant 1; the Pauli matrices σ (times i) form a basis of its algebra.
- Lorentz group $SO(3, 1)$; rotation group $SO(3)$: $R^\top R = \mathbb{K}$, $\det R = 1$. $SO(3)$ is a subgroup of $SO(3, 1)$.
- V^μ is an irreducible representation of $SO(3, 1)$ but reducible under $SO(3)$ (scalar $\oplus$ vector).

Particles are identified with irreducible representations of the Poincaré group (Lorentz transformations and translations).

- A **group** is a set of elements $\{g_i\}$ with a composition rule $\cdot$ such that $g_i \cdot g_j = g_k$ (plus associativity, identity and inverses).
- A **representation** of a group is a map (embedding) of the group elements g_i into operators that act on a vector space, preserving the composition rule. For finite-dimensional representations the elements can always be mapped to matrices.
- A representation is **irreducible** if there is no proper subspace that transforms among itself.

8.2 The Lorentz group $SO(3, 1)$

A set of objects ϕ^i , $i = 1, \dots, n$, transforms in a representation R of dimension n if under a Lorentz transformation

$$\phi^i \rightarrow \left[e^{-\frac{i}{2} \omega_{\mu\nu} J_R^{\mu\nu}} \right]^i_j \phi^j,$$

where the $J_R^{\mu\nu}$ ($n \times n$ matrices) are the Lorentz generators in the representation R , i.e. $\delta\phi^i = -\frac{i}{2} \omega_{\mu\nu} (J_R^{\mu\nu})^i_j \phi^j$; (μ, ν) label the generators and (i, j) are matrix indices.

- Scalar field: $\phi'(x') = \phi(x)$, $i, j \in \{1\}$, $J_{\text{scalar}}^{\mu\nu} = 0$: the trivial representation (the change of $\phi(x)$ comes only from the argument, $\delta\phi = -\omega^\mu_\nu x^\nu \partial_\mu \phi$; with $\Lambda = e^{-\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu}}$ and $J^{\mu\nu} = -J^{\nu\mu}$ there are 6 independent generators).
- Four-vector V^μ : the indices i, j are themselves Lorentz indices, and

$$(J^{\mu\nu})^\rho_\sigma = i(\eta^{\mu\rho} \delta^\nu_\sigma - \eta^{\nu\rho} \delta^\mu_\sigma).$$

The generators satisfy the Lie algebra of $SO(3, 1)$:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(\eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}).$$

Writing $J^i = \frac{1}{2} \varepsilon^{ijk} J^{jk}$ and $K^i = J^{i0}$, so that $\Lambda = e^{i\vec{\theta} \cdot \vec{J} + i\vec{\beta} \cdot \vec{K}}$,

$$\begin{aligned} [J^i, J^j] &= i\varepsilon^{ijk} J^k & (SO(3) \simeq SU(2)), \\ [J^i, K^j] &= i\varepsilon^{ijk} K^k & (\vec{K} \text{ is a vector}), \\ [K^i, K^j] &= -i\varepsilon^{ijk} J^k. \end{aligned}$$

8.3 $SU(2)$ and spinors

$SU(2)$ is the group of unitary matrices with determinant 1; it is periodic only under rotations by 4π ($SU(2)$ double-covers $SO(3)$). Its representations are labelled by $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, and the spin- j representation has $2j + 1$ dimensions. The spinorial (fundamental) representation $j = \frac{1}{2}$ is two dimensional,

$$J^i = \frac{\sigma^i}{2}, \quad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma^i \sigma^j = \delta^{ij} + i\varepsilon^{ijk} \sigma^k.$$

All other representations are built from tensor products of spin- $\frac{1}{2}$ representations, e.g. $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$ (dimensions $2 \times 2 = 1 + 3$).

Combining the $SO(3, 1)$ generators as

$$\vec{J}_\pm = \frac{\vec{J} \pm i\vec{K}}{2} \implies [J_\pm^i, J_\pm^j] = i\varepsilon^{ijk} J_\pm^k, \quad [J_+^i, J_-^j] = 0:$$

the Lorentz algebra is composed of two (commuting) copies of the $SU(2)$ algebra. Representations of the Lorentz group can therefore be labelled by (j_+, j_-) , with dimension $(2j_+ + 1)(2j_- + 1)$, rotation generators $\vec{J} = \vec{J}_+ + \vec{J}_-$, and spins from $|j_+ - j_-|$ to $j_+ + j_-$.

Spin- $\frac{1}{2}$ particles are associated with $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$, the **Weyl spinors**: one has $\vec{J} = \vec{\sigma}/2$, $\vec{K} = i\vec{\sigma}/2$ (left-handed spinor ψ_L) and the other $\vec{J} = \vec{\sigma}/2$, $\vec{K} = -i\vec{\sigma}/2$ (right-handed spinor ψ_R):

$$\begin{aligned} \psi_L \rightarrow \Lambda_L \psi_L &= e^{\frac{1}{2} (i\vec{\theta} - \vec{\beta}) \cdot \vec{\sigma}} \psi_L, & \psi_R \rightarrow \Lambda_R \psi_R &= e^{\frac{1}{2} (i\vec{\theta} + \vec{\beta}) \cdot \vec{\sigma}} \psi_R, \\ \delta\psi_L &= \frac{1}{2} (i\theta_i - \beta_i) \sigma^i \psi_L, & \delta\psi_R &= \frac{1}{2} (i\theta_i + \beta_i) \sigma^i \psi_R. \end{aligned}$$

8.4 Lorentz-invariant Lagrangian for Weyl and Dirac spinors

To have a Lorentz-invariant mass term one needs to combine ψ_L and ψ_R : $m(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L)$. The kinetic terms are $i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L + i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R$, with $\sigma^\mu := (\mathbb{1}, \sigma^i)$, $\bar{\sigma}^\mu := (\mathbb{1}, -\sigma^i)$ ($\psi_L^\dagger \bar{\sigma}^\mu \psi_L$ is a four-vector). Hence

$$\mathcal{L} = i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L + i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R - m(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L).$$

For $m \rightarrow 0$: $\sigma^\mu \partial_\mu \psi_R = 0$ and $\bar{\sigma}^\mu \partial_\mu \psi_L = 0$, i.e.

$$\frac{\vec{\sigma} \cdot \vec{p}}{|\vec{p}|} \psi_R = \psi_R, \quad \frac{\vec{\sigma} \cdot \vec{p}}{|\vec{p}|} \psi_L = -\psi_L :$$

ψ_R and ψ_L are eigenvectors of the **helicity operator** $h := \vec{\sigma} \cdot \vec{p}/|\vec{p}|$ with eigenvalues $+1$ and -1 .

Dirac equation. The four-component **Dirac spinor**, the gamma matrices and the Dirac adjoint are

$$\psi := \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad \gamma^\mu := \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \quad \bar{\psi} := \psi^\dagger \gamma^0 = (\psi_R^\dagger, \psi_L^\dagger),$$

and

$$\mathcal{L}_D = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi = \bar{\psi}(i\not{\partial} - m)\psi \implies (i\gamma^\mu \partial_\mu - m)\psi = 0,$$

the **Dirac equation**. Applying $(-i\not{\partial} - m)$ to it yields the Klein–Gordon equation, $(\partial_\mu \partial^\mu + m^2)\psi = 0$. The conjugate momentum is $\pi_\psi = i\psi^\dagger$, so there are 4 degrees of freedom. $\bar{\psi}\psi$ is a scalar and $\bar{\psi}\gamma^\mu\psi$ a vector.

Lorentz transformation of Dirac spinors.

$$\Lambda_V = e^{i\theta_j J^j + i\beta_j K^j}, \quad \Lambda_S = \begin{pmatrix} \Lambda_L & 0 \\ 0 & \Lambda_R \end{pmatrix} = e^{-\frac{i}{2}\omega_{\rho\sigma} S^{\rho\sigma}}, \quad S^{\rho\sigma} := \frac{i}{4}[\gamma^\rho, \gamma^\sigma],$$

with $\omega_{ij} = -\varepsilon_{ijk}\theta_k$, $\omega_{0i} = -\beta_i$, and

$$S^{0i} = -\frac{i}{2} \begin{pmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{pmatrix}, \quad S^{ij} = \frac{1}{2} \varepsilon^{ijk} \begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}.$$

The field also receives a contribution from the change of coordinates:

$$\delta\psi^\alpha = -\omega^\mu{}_\nu x^\nu \partial_\mu \psi^\alpha - \frac{i}{2} \omega_{\rho\sigma} (S^{\rho\sigma})^\alpha{}_\beta \psi^\beta.$$

9 Fermions II

9.1 Noether currents

- Translations, $x'^\mu = x^\mu + c^\mu$, $\delta\psi = -c^\mu \partial_\mu \psi$:

$$T^{\mu\nu} = i\bar{\psi}\gamma^\mu \partial^\nu \psi, \quad E = \int d^3x T^{00} = \int d^3x \bar{\psi}(-i\gamma^i \partial_i + m)\psi.$$

- Lorentz, $x'^\mu = \Lambda^\mu{}_\nu x^\nu = (\delta^\mu_\nu + \omega^\mu{}_\nu)x^\nu$:

$$(\mathcal{J}^\mu)^{\rho\sigma} = (x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho}) + \bar{\psi}\gamma^\mu S^{\rho\sigma}\psi.$$

9.2 Quantisation of the Dirac field

Starting from $(i\gamma^\mu \partial_\mu - m)\psi = 0$, the ansatz $\psi(x) = u(p)e^{-ip \cdot x}$ (positive-frequency solutions) gives

$$(\gamma^\mu p_\mu - m)u(p) = 0 \quad \implies \quad \begin{pmatrix} -m & p \cdot \sigma \\ p \cdot \bar{\sigma} & -m \end{pmatrix} u(p) = 0.$$

Using $(p \cdot \sigma)(p \cdot \bar{\sigma}) = p^2 = m^2$,

$$u(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi \\ \sqrt{p \cdot \bar{\sigma}} \xi \end{pmatrix}, \quad \xi^\dagger \xi = 1.$$

Similarly the negative-frequency solutions $\psi(x) = v(p)e^{ip \cdot x}$ have

$$v(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \eta \\ -\sqrt{p \cdot \bar{\sigma}} \eta \end{pmatrix}, \quad \eta^\dagger \eta = 1.$$

In general the basis chosen is the eigenbasis of σ^3 : $\xi_\uparrow = \eta_\uparrow = (1, 0)^\top$, $\xi_\downarrow = \eta_\downarrow = (0, 1)^\top$.

- Rest frame, $p^\mu = (m, 0, 0, 0)$:

$$u_\uparrow = \sqrt{m} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_\downarrow = \sqrt{m} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_\uparrow = \sqrt{m} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad v_\downarrow = \sqrt{m} \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}.$$

- Boost in the z direction:

$$u_\uparrow = \begin{pmatrix} \sqrt{E - p_3} \\ 0 \\ \sqrt{E + p_3} \\ 0 \end{pmatrix}, \quad u_\downarrow = \begin{pmatrix} 0 \\ \sqrt{E + p_3} \\ 0 \\ \sqrt{E - p_3} \end{pmatrix}, \quad v_\uparrow = \begin{pmatrix} \sqrt{E - p_3} \\ 0 \\ -\sqrt{E + p_3} \\ 0 \end{pmatrix}, \quad v_\downarrow = \begin{pmatrix} 0 \\ \sqrt{E + p_3} \\ 0 \\ -\sqrt{E - p_3} \end{pmatrix}.$$

- Massless limit $m \rightarrow 0$ ($p_3 = E$):

$$u_\uparrow = \sqrt{2E} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_\downarrow = \sqrt{2E} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_\uparrow = \sqrt{2E} \begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad v_\downarrow = \sqrt{2E} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

These are eigenstates of the helicity operator $h = \frac{1}{2}\varepsilon_{ijk}\hat{p}^i S^{jk} = \frac{1}{2}\hat{p}_i \begin{pmatrix} \sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix}$: spin-up and spin-down Weyl spinors.

In general the helicity of a particle is frame dependent; for massless particles it is invariant.

Canonical anticommutation relations. With $\pi_\psi = i\psi^\dagger$, one could naively postulate $[\psi_\alpha(\vec{x}), \psi_\beta^\dagger(\vec{y})] = \delta^{(3)}(\vec{x} - \vec{y})\delta_{\alpha\beta}$, but this runs into trouble. The correct way to quantise ψ is to impose

$$\{\psi_\alpha(\vec{x}), \psi_\beta^\dagger(\vec{y})\} = \delta^{(3)}(\vec{x} - \vec{y})\delta_{\alpha\beta}.$$

Then

$$\begin{aligned} \psi(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \sum_{s=1}^2 \left(b_{\vec{p}}^s u^s(\vec{p}) e^{i\vec{p} \cdot \vec{x}} + c_{\vec{p}}^{s\dagger} v^s(\vec{p}) e^{-i\vec{p} \cdot \vec{x}} \right), \\ \psi^\dagger(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \sum_{s=1}^2 \left(b_{\vec{p}}^{s\dagger} u^{s\dagger}(\vec{p}) e^{-i\vec{p} \cdot \vec{x}} + c_{\vec{p}}^s v^{s\dagger}(\vec{p}) e^{i\vec{p} \cdot \vec{x}} \right), \end{aligned}$$

$$\{b_{\vec{p}}^s, b_{\vec{q}}^{r\dagger}\} = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \delta^{sr}, \quad \{c_{\vec{p}}^s, c_{\vec{q}}^{r\dagger}\} = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \delta^{sr}.$$

The Hamiltonian is

$$H = \int d^3x (\pi_\psi \dot{\psi} - \mathcal{L}) = \int d^3x \bar{\psi} (-i\gamma^i \partial_i + m) \psi,$$

$$:H: = \int \frac{d^3p}{(2\pi)^3} \sum_s E_{\vec{p}} (b_{\vec{p}}^{s\dagger} b_{\vec{p}}^s + c_{\vec{p}}^{s\dagger} c_{\vec{p}}^s), \quad H = :H: - (2\pi)^3 \delta^{(3)}(0) \dots,$$

where the vacuum energy has the opposite sign compared with the bosonic system; this could solve the problem of the infinite energy (supersymmetry). Also $[H, b_{\vec{p}}^{s\dagger}] = E_{\vec{p}} b_{\vec{p}}^{s\dagger}$ and $[H, c_{\vec{p}}^{s\dagger}] = E_{\vec{p}} c_{\vec{p}}^{s\dagger}$. (Next time: spin–statistics.)

9.3 Internal symmetries

- $\psi \rightarrow e^{i\alpha} \psi$ gives the **vector current** $j^\mu = \bar{\psi} \gamma^\mu \psi$, with $Q = \int d^3x \psi^\dagger \psi$ and

$$:Q: = \int \frac{d^3p}{(2\pi)^3} \sum_s [b_{\vec{p}}^{s\dagger} b_{\vec{p}}^s - c_{\vec{p}}^{s\dagger} c_{\vec{p}}^s] :$$

fermions and antifermions have opposite charge.

- For $m \rightarrow 0$, **chiral symmetry**: $\psi \rightarrow e^{i\alpha\gamma^5} \psi$ with $\gamma^5 := i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -\mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix}$, giving the **axial current** $j_A^\mu = \bar{\psi} \gamma^\mu \gamma^5 \psi$, with $\partial_\mu j_A^\mu = 2im\bar{\psi} \gamma^5 \psi = 0$ for $m = 0$.

9.4 Feynman propagator for fermions

In the Heisenberg picture,

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \sum_{s=1}^2 (b_{\vec{p}}^s u^s(p) e^{-ip \cdot x} + c_{\vec{p}}^{s\dagger} v^s(p) e^{ip \cdot x}),$$

and

$$S_F(x - y) = \langle 0 | T \{ \psi(x) \bar{\psi}(y) \} | 0 \rangle, \quad T \{ \psi(x) \bar{\psi}(y) \} = \begin{cases} \psi(x) \bar{\psi}(y), & x^0 > y^0, \\ -\bar{\psi}(y) \psi(x), & x^0 < y^0. \end{cases}$$

Using $\sum_s u^s \bar{u}^s = \not{p} + m$ and $\sum_s v^s \bar{v}^s = \not{p} - m$,

$$\begin{aligned} \langle 0 | \psi(x) \bar{\psi}(y) | 0 \rangle &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} (\not{p} + m) e^{-ip \cdot (x-y)}, \\ \langle 0 | \bar{\psi}(y) \psi(x) | 0 \rangle &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} (\not{p} - m) e^{ip \cdot (x-y)}, \end{aligned}$$

$$S_F(x - y) = \int \frac{d^4p}{(2\pi)^4} \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x-y)} = (i\not{\partial}_x + m) \Delta_F(x - y), \quad S_F(p) = \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}.$$

9.5 Yukawa theory with fermions

(Peskin–Schroeder 4.7.)

$$\mathcal{L}_Y = \frac{1}{2} (\partial\phi)^2 - \frac{\mu^2}{2} \phi^2 + \bar{\psi} (i\not{\partial} - m) \psi - g \phi \bar{\psi} \psi.$$

Feynman rules (momentum space):

- incoming fermion: $u^s(p)$; outgoing fermion: $\bar{u}^s(p)$; incoming antifermion: $\bar{v}^s(p)$; outgoing antifermion: $v^s(p)$;
- vertex: $-ig$, with momentum conservation;
- integrate over undetermined internal momenta;
- propagators: $\frac{i}{p^2 - \mu^2 + i\epsilon}$ (scalar), $\frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}$ (fermion);
- a minus sign for every closed fermion loop, for every open fermion line that contains an outgoing antifermion, and for every crossing of fermions.

Example: $\psi\psi \rightarrow \psi\psi$ at order g^2 . With $H_I = g\bar{\psi}\psi\phi$,

$$\langle \vec{p}_3 \vec{p}_4 | S | \vec{p}_1 \vec{p}_2 \rangle = \frac{(-ig)^2}{2!} \langle \vec{p}_3 \vec{p}_4 | T \left(\int d^4x \bar{\psi}_I \psi_I \phi_I(x) \int d^4y \bar{\psi}_I \psi_I \phi_I(y) \right) | \vec{p}_1 \vec{p}_2 \rangle,$$

with $\langle \vec{p}_3 \vec{p}_4 | \propto \langle 0 | b_{\vec{p}_4} b_{\vec{p}_3}$ and $|\vec{p}_1 \vec{p}_2 \rangle \propto b_{\vec{p}_1}^\dagger b_{\vec{p}_2}^\dagger | 0 \rangle$. The factor $2!$ from $x \leftrightarrow y$ cancels the $\frac{1}{2!}$ of the Dyson formula (no further combinatorial factors). Two contractions contribute:

$$\begin{aligned} i\mathcal{A}_t &= \frac{ig^2}{(p_1 - p_3)^2 - \mu^2 + i\epsilon} [\bar{u}^{s_3}(p_3) u^{s_1}(p_1)] [\bar{u}^{s_4}(p_4) u^{s_2}(p_2)], \\ i\mathcal{A}_u &= \frac{-ig^2}{(p_1 - p_4)^2 - \mu^2 + i\epsilon} [\bar{u}^{s_4}(p_4) u^{s_1}(p_1)] [\bar{u}^{s_3}(p_3) u^{s_2}(p_2)], \end{aligned}$$

with a relative minus sign from the exchange of identical fermions.

Fermion loops. For the diagram with a closed fermion loop and four external scalars,

$$\begin{aligned} \sum_{abcd} \bar{\psi}_a(x_1) \psi_a(x_1) \bar{\psi}_b(x_2) \psi_b(x_2) \bar{\psi}_c(x_3) \psi_c(x_3) \bar{\psi}_d(x_4) \psi_d(x_4) \\ = -\text{Tr}[S_F(x_4 - x_1) S_F(x_1 - x_2) S_F(x_2 - x_3) S_F(x_3 - x_4)]. \end{aligned}$$

t -channel of $\psi\bar{\psi}$ annihilation into scalars.

$$i\mathcal{A}_t = \frac{-ig^2}{(p_1 - p_3)^2 - m^2 + i\epsilon} \bar{v}^{s_2}(p_2) (\not{p}_1 - \not{p}_3 + m) u^{s_1}(p_1).$$

Squared amplitudes. For $d\sigma/d\Omega \propto |\mathcal{A}|^2$,

$$\mathcal{A}\mathcal{A}^* = [\bar{u}(p)\Gamma_1 u(k)][\bar{u}(p)\Gamma_2 u(k)]^*, \quad [\bar{u}(p)\Gamma_2 u(k)]^* = \bar{u}(k)\gamma^0\Gamma_2^\dagger\gamma^0 u(p),$$

and summing over spins with $\sum_r u^r(p)\bar{u}^r(p) = \not{p} + m$,

$$\sum_{\text{spins}} [\bar{u}(p)\Gamma_1 u(k)][\bar{u}(p)\Gamma_2 u(k)]^* = \text{Tr}\{(\not{p} + m)\Gamma_1(\not{k} + m)\gamma^0\Gamma_2^\dagger\gamma^0\}.$$

9.6 Gauge fields

Maxwell Lagrangian (spin-one field):

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \quad F_{\mu\nu} := \partial_\mu A_\nu - \partial_\nu A_\mu,$$

invariant under $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$ (**gauge invariance**). The equations of motion are

$$\square A_\mu - \partial_\mu \partial_\nu A^\nu = 0, \quad \text{i.e.} \quad -\nabla^2 A_0 - \partial_0 \partial_i A^i = 0, \quad \square A_i - \partial_i (\partial_0 A_0 - \partial_j A^j) = 0.$$

Fixing the gauge.

- Coulomb gauge, $\partial_i A_i = 0$ (not completely fixed): then $\nabla^2 A_0 = 0$, and with the residual freedom $A_0 \rightarrow A_0 + \partial_0 \alpha$ we set $A_0 = 0$, so $\square A_i = 0$: $\vec{A}$ has just two degrees of freedom.
- Lorenz gauge, $\partial_\mu A^\mu = 0$, with additional gauge freedom $\partial_\mu \partial^\mu \alpha = 0$.

Quantisation of A in the Coulomb gauge.

$$\pi^0 = \frac{\partial \mathcal{L}}{\partial \dot{A}_0} = 0, \quad \pi^i = \frac{\partial \mathcal{L}}{\partial \dot{A}_i} = -F^{0i} = E^i,$$

$$H = \int d^3x [\pi^i \dot{A}_i - \mathcal{L}] = \int d^3x \left[\frac{1}{2} (\vec{E}^2 + \vec{B}^2) - A_0 (\nabla \cdot \vec{E}) \right],$$

where A_0 acts as a Lagrange multiplier. With $\square \vec{A} = 0$ and $\nabla \cdot \vec{A} = 0$ (transverse), $\vec{A}(x) = \vec{\varepsilon}(p) e^{-ip \cdot x}$ with $\vec{p} \cdot \vec{\varepsilon} = 0$: $\vec{\varepsilon}$ is the **polarisation vector**. For $\vec{p} = (0, 0, p)$, circular polarisations are $\vec{\varepsilon}_{1,2} = \frac{1}{\sqrt{2}}(1, \pm i, 0)$. Then

$$\begin{aligned} \vec{A}(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \sum_{\lambda} \left(\vec{\varepsilon}_{\lambda}(\vec{p}) a_{\vec{p}}^{\lambda} e^{i\vec{p} \cdot \vec{x}} + \vec{\varepsilon}_{\lambda}^*(\vec{p}) a_{\vec{p}}^{\lambda\dagger} e^{-i\vec{p} \cdot \vec{x}} \right), \\ \vec{E}(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} (-i) \sqrt{\frac{E_{\vec{p}}}{2}} \sum_{\lambda} \left(\vec{\varepsilon}_{\lambda}(\vec{p}) a_{\vec{p}}^{\lambda} e^{i\vec{p} \cdot \vec{x}} - \text{h.c.} \right), \end{aligned}$$

with $[a_{\vec{p}}^{\lambda}, a_{\vec{p}'}^{\lambda'\dagger}] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{p}') \delta^{\lambda\lambda'}$, so that

$$[A^i(t, \vec{x}), E^j(t, \vec{y})] = i \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p} \cdot (\vec{x} - \vec{y})} \left(\delta^{ij} - \frac{p^i p^j}{|\vec{p}|^2} \right) =: i \delta_{\perp}^{ij}(\vec{x} - \vec{y}), \quad \sum_{\lambda} \varepsilon_{\lambda}^i \varepsilon_{\lambda}^{j*} = \delta^{ij} - \frac{p^i p^j}{|\vec{p}|^2}.$$

Then

$$:H: = \int \frac{d^3p}{(2\pi)^3} \sum_{\lambda} E_{\vec{p}} a_{\vec{p}}^{\lambda\dagger} a_{\vec{p}}^{\lambda}, \quad :\vec{P}: = \int \frac{d^3p}{(2\pi)^3} \sum_{\lambda} \vec{p} a_{\vec{p}}^{\lambda\dagger} a_{\vec{p}}^{\lambda}, \quad E_{\vec{p}} = |\vec{p}|,$$

and the propagator in the Coulomb gauge is

$$\Delta_{\perp}^{ij}(x - y) := \langle 0 | T \{ A^i(x) A^j(y) \} | 0 \rangle = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 + i\epsilon} \left(\delta^{ij} - \frac{p^i p^j}{|\vec{p}|^2} \right) e^{-ip \cdot (x - y)},$$

while in the Lorenz (Feynman) gauge $\Delta_F^{\mu\nu}(p) = \frac{-i\eta^{\mu\nu}}{p^2 + i\epsilon}$.

Coupling to a current. With $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - A_{\mu} J^{\mu} + \mathcal{L}_m$, in the Coulomb gauge

$$\mathcal{L} = \frac{1}{2} \dot{A}_i^2 - \frac{1}{2} (\partial_i A_j)^2 - \frac{1}{2} A_0 \nabla^2 A_0 - A_0 J^0 + A_i J^i.$$

Now $\nabla^2 A_0 = -J^0$, so A_0 cannot be set to zero this time:

$$A_0(\vec{x}) = \int d^3y \frac{J^0(\vec{y})}{4\pi|\vec{x} - \vec{y}|}, \quad \mathcal{L} = \frac{1}{2} (\partial A_i)^2 + A_i J^i + \frac{1}{2} \int d^3y \frac{J^0(\vec{x}) J^0(\vec{y})}{4\pi|\vec{x} - \vec{y}|},$$

where the last is a non-local term. Correspondingly

$$\Delta_F^{00}(x - y) = \frac{\delta(x^0 - y^0)}{4\pi|\vec{x} - \vec{y}|} = \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip \cdot (x - y)}}{|\vec{p}|^2}, \quad \Delta_F^{\mu\nu}(p) = \begin{cases} i/|\vec{p}|^2, & \mu = \nu = 0, \\ \Delta_{\perp}^{ij}(p), & \mu, \nu = i, j, \\ 0, & \text{otherwise,} \end{cases}$$

i.e. $\Delta_F^{\mu\nu}(p) = \frac{i}{p^2 + i\epsilon} \left(P_{\perp}^{\mu\nu} + \frac{p^2}{|\vec{p}|^2} n^\mu n^\nu \right)$ with $n^\mu = (1, 0, 0, 0)$, $P_{\perp}^{00} = P_{\perp}^{0i} = P_{\perp}^{i0} = 0$, $P_{\perp}^{ij} = \delta^{ij} - p^i p^j / |\vec{p}|^2$. Contracted with conserved currents the extra terms drop out:

$$J_\mu^*(p) \Delta_F^{\mu\nu}(p) J_\nu(p) = \frac{-i}{p^2} J_\mu^*(p) J^\mu(p) \implies \Delta_F^{\mu\nu}(p) = \frac{-i\eta^{\mu\nu}}{p^2 + i\epsilon}.$$

9.7 Discrete symmetries for fermions

C (charge conjugation), P (parity), T (time reversal).

Statement (CPT theorem). All relativistic quantum field theories are invariant under the CPT transformation.

- Charge conjugation (particles $\rightarrow$ antiparticles): $C b_p^s C = c_p^s$, $C c_p^s C = b_p^s$, so $\psi \rightarrow -i\gamma^2 \psi^*$.
- Parity ($\vec{x} \rightarrow -\vec{x}$): $P b_p^s P = b_{-p}^s$, $P c_p^s P = -c_{-p}^s$, so $\psi(t, \vec{x}) \rightarrow \gamma^0 \psi(t, -\vec{x})$ (exchanging $\psi_L \leftrightarrow \psi_R$).
- Time reversal ($t \rightarrow -t$, $\partial_t \rightarrow -\partial_t$): T must be antilinear, $Tz = z^*T$, and it reverses the momentum (under complex conjugation $\gamma^{0,1,3} \rightarrow \gamma^{0,1,3}$, $\gamma^2 \rightarrow -\gamma^2$): $T b_p^s T = b_{-p}^{-s}$, $T c_p^s T = c_{-p}^{-s}$, so $\psi(t, \vec{x}) \rightarrow \gamma^1 \gamma^3 \psi(-t, \vec{x})$.
- CPT: $\psi(x) \rightarrow -\gamma^5 \psi^*(-x)$.

10 Quantum Electrodynamics

(“The strange theory of light and matter”).

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - A_\mu J^\mu, \quad \partial_\mu F^{\mu\nu} = J^\nu \implies \partial_\nu J^\nu = 0 :$$

current conservation. Under $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$, $\delta S = \int d^4x \alpha \partial_\mu J^\mu$: the action is gauge invariant if $\partial_\mu J^\mu = 0$ (if J^μ is conserved).

Gauging the phase symmetry of fermions. Phase rotations $\psi \rightarrow e^{i\beta} \psi$ are a symmetry. What if this transformation were local, $\psi \rightarrow e^{i\beta(x)} \psi$? $\bar{\psi}\psi$ is still invariant, but

$$\partial_\mu \psi \rightarrow \partial_\mu (e^{i\beta(x)} \psi) = e^{i\beta(x)} [\partial_\mu \psi + i(\partial_\mu \beta) \psi], \quad \delta S_{\text{Dirac}} = - \int d^4x (\partial_\mu \beta) \bar{\psi} \gamma^\mu \psi.$$

Let us redefine $\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$ (the **minimal coupling** procedure). Under

$$A_\mu \rightarrow A_\mu + \partial_\mu \alpha, \quad \psi \rightarrow e^{i\beta(x)} \psi,$$

the action is invariant if $\beta = -q\alpha$. Then

$$\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi = \bar{\psi}(i\not{\partial} - m)\psi - qA_\mu \underbrace{\bar{\psi} \gamma^\mu \psi}_{j^\mu}, \quad :Q: = q \int \frac{d^3p}{(2\pi)^3} \sum_s [b_p^{s\dagger} b_p^s - c_p^{s\dagger} c_p^s],$$

where q is the electric charge. The **fine-structure constant** $\alpha = e^2/4\pi \simeq 1/137$ is a small number.

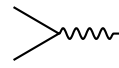
Scalar QED. Similarly, for a complex scalar

$$\begin{aligned} \partial_\mu \varphi^* \partial^\mu \varphi - m^2 \varphi^* \varphi &\longrightarrow D_\mu \varphi (D^\mu \varphi)^* - m^2 \varphi^* \varphi \\ &= \partial_\mu \varphi \partial^\mu \varphi^* - m^2 \varphi^* \varphi + ieA_\mu \underbrace{(\varphi \partial^\mu \varphi^* - \varphi^* \partial^\mu \varphi)}_{j^\mu} + e^2 \varphi \varphi^* A_\mu A^\mu. \end{aligned}$$

QED Lagrangian.

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\not{D} - m)\psi = \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\not{D} - m)\psi}_{\mathcal{L}_0} \underbrace{-qA_\mu\bar{\psi}\gamma^\mu\psi}_{\mathcal{L}_{\text{int}}}.$$

Feynman rules (momentum space):

	$\frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}$	fermion propagator
	$\frac{-i\eta_{\mu\nu}}{p^2 + i\epsilon}$	photon propagator
	$-ie\gamma^\mu$	QED vertex
external lines	$u^s(p), \bar{u}^s(p)$	fermion in, out
	$\bar{v}^s(p), v^s(p)$	antifermion in, out
	$\varepsilon_\mu^\lambda(p), \varepsilon_\mu^{\lambda*}(p)$	photon in, out

10.1 Some processes

$$e^-e^- \rightarrow e^-e^-.$$

$$\begin{aligned} i\mathcal{A}_t &= \bar{u}^{s_3}(p_3)(-ie\gamma^\mu)u^{s_1}(p_1) \frac{-i\eta_{\mu\nu}}{(p_1 - p_3)^2} \bar{u}^{s_4}(p_4)(-ie\gamma^\nu)u^{s_2}(p_2) \\ &= \frac{ie^2 [\bar{u}^{s_3}(p_3)\gamma^\mu u^{s_1}(p_1)] [\bar{u}^{s_4}(p_4)\gamma_\mu u^{s_2}(p_2)]}{(p_1 - p_3)^2}, \\ i\mathcal{A}_u &= \frac{-ie^2 [\bar{u}^{s_4}(p_4)\gamma^\mu u^{s_1}(p_1)] [\bar{u}^{s_3}(p_3)\gamma_\mu u^{s_2}(p_2)]}{(p_1 - p_4)^2}. \end{aligned}$$

$$e^-e^+ \rightarrow \gamma\gamma.$$

$$i\mathcal{A} = -ie^2 \bar{v}^{s_2}(p_2)\gamma^\mu \varepsilon_\mu^*(p_4) \frac{\not{p}_1 - \not{p}_3 + m}{(p_1 - p_3)^2 - m^2} \gamma^\nu u^{s_1}(p_1) \varepsilon_\nu^*(p_3) + (3 \leftrightarrow 4).$$

$e^-e^+ \rightarrow e^-e^+$. t -channel ($q = p_1 - p_3$) and s -channel ($q = p_1 + p_2$):

$$\begin{aligned} t: & \quad \bar{u}^{s_3}(p_3)(-ie\gamma^\mu)u^{s_1}(p_1) \frac{-i\eta_{\mu\nu}}{(p_1 - p_3)^2} \bar{v}^{s_2}(p_2)(-ie\gamma^\nu)v^{s_4}(p_4), \\ s: & \quad \frac{ie^2 [\bar{u}^{s_3}(p_3)\gamma^\mu v^{s_4}(p_4)] [\bar{v}^{s_2}(p_2)\gamma_\mu u^{s_1}(p_1)]}{(p_1 + p_2)^2}. \end{aligned}$$

$e^-\mu^- \rightarrow e^-\mu^-$. Only the t -channel, $q = p_1 - p_3$:

$$i\mathcal{A}_t = \frac{ie^2}{q^2} [\bar{u}_e(p_3)\gamma^\mu u_e(p_1)] [\bar{u}_\mu(p_4)\gamma_\mu u_\mu(p_2)], \quad |\mathcal{A}|^2 = \frac{e^4}{q^4} L_{(e)}^{\mu\nu} L_{(\mu)}_{\mu\nu},$$

with $L_{(e)}^{\mu\nu} = [\bar{u}_e(p_3)\gamma^\mu u_e(p_1)][\bar{u}_e(p_3)\gamma^\nu u_e(p_1)]^*$ and similarly for the muon. Averaging over initial and summing over final spins, with $\sum_s u^s(p)\bar{u}^s(p) = \not{p} + m$,

$$\frac{1}{4} \sum_{\text{spins}} |\mathcal{A}|^2 = \frac{e^4}{4q^4} \text{Tr}[\gamma^\mu(\not{p}_1 + m_e)\gamma^\nu(\not{p}_3 + m_e)] \text{Tr}[\gamma_\mu(\not{p}_2 + m_\mu)\gamma_\nu(\not{p}_4 + m_\mu)].$$

Using the trace identities

$$\text{Tr}\{\not{a}\not{b}\} = 4a \cdot b, \quad \text{Tr}\{\not{a}\not{b}\not{c}\not{d}\} = 4(a \cdot b \cdot c \cdot d - a \cdot c \cdot b \cdot d + a \cdot d \cdot b \cdot c), \quad \text{Tr}\{\text{odd number of } \gamma\text{'s}\} = 0,$$

$$\frac{1}{4} \sum_{\text{spins}} |\mathcal{A}|^2 = \frac{2e^4}{t^2} [s^2 + u^2 - 4(m_e^2 + m_\mu^2)(s + u) + 6(m_e^2 + m_\mu^2)^2].$$

In the high-energy limit $s, u \gg m_e^2, m_\mu^2$, with $d\sigma/d\Omega = |\mathcal{A}|^2/64\pi^2 s$,

$$\frac{d\sigma}{d\Omega} = \frac{e^4}{32\pi^2 s} \frac{s^2 + u^2}{t^2}.$$

Part II

Path Integrals and Renormalisation

About this part. This part follows Srednicki's *Quantum Field Theory* and complements Part I (canonical quantisation, Feynman rules for ϕ^4 , Yukawa theory, fermions and QED) with the path-integral formulation and the renormalisation of a scalar theory. To stay close to Srednicki, we switch conventions here:

- **Metric:** mostly plus, $g^{\mu\nu} = \text{diag}(-1, +1, +1, +1)$, so that $k^2 = -\omega^2 + \vec{k}^2$ and an on-shell particle has $k^2 = -m^2$.
- **Free Lagrangian:** $\mathcal{L}_0 = -\frac{1}{2}\partial^\mu\phi\partial_\mu\phi - \frac{1}{2}m^2\phi^2$, identical to Part I up to the sign flip of the metric.
- **Propagator:** $\Delta(x-y) := i\langle 0|T\phi(x)\phi(y)|0\rangle$, i.e. $\Delta = i\Delta_F$ in terms of the Feynman propagator of Part I, with $\tilde{\Delta}(k^2) = 1/(k^2 + m^2 - i\epsilon)$.
- **Lorentz-invariant measure:** $\widetilde{d}k := \frac{d^3k}{(2\pi)^3 2\omega}$, $\omega = \sqrt{\vec{k}^2 + m^2}$.

We keep $\hbar$ explicit only where it helps (the path integral and the loop expansion).

11 Poincaré Symmetry and the Free Field Revisited

11.1 The Poincaré algebra

A Lorentz transformation Λ acts on Hilbert space through a unitary operator $U(\Lambda)$. For an infinitesimal transformation $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \delta\omega^\mu{}_\nu$ (with $\delta\omega_{\mu\nu}$ antisymmetric, the parameters),

$$U(\Lambda) = \mathbb{1} + \frac{i}{2\hbar}\delta\omega_{\mu\nu}M^{\mu\nu}, \quad U^\dagger(\Lambda)M^{\mu\nu}U(\Lambda) = \Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta M^{\alpha\beta},$$

where the $M^{\mu\nu} = -M^{\nu\mu}$ are the generators. They satisfy the Lorentz algebra

$$[M^{\mu\nu}, M^{\rho\sigma}] = i\hbar(g^{\mu\rho}M^{\nu\sigma} - g^{\nu\rho}M^{\mu\sigma} - g^{\mu\sigma}M^{\nu\rho} + g^{\nu\sigma}M^{\mu\rho}).$$

One could start the theory from this algebra (it is not restricted to four-dimensional matrices; U is a unitary representation), or from the defining condition $\Lambda^\mu_\alpha \Lambda^\nu_\beta g_{\mu\nu} = g_{\alpha\beta}$ (restricted to 4×4 matrices). Defining the angular momentum and boost generators

$$J_i := \frac{1}{2} \varepsilon_{ijk} M^{jk}, \quad K_i := M^{i0},$$

the Lorentz algebra reads

$$[J_i, J_j] = i\hbar \varepsilon_{ijk} J_k, \quad [J_i, K_j] = i\hbar \varepsilon_{ijk} K_k, \quad [K_i, K_j] = -i\hbar \varepsilon_{ijk} J_k.$$

The rotations close among themselves; the boosts do not (two boosts produce a rotation). For instance, $\delta\omega_{\mu\nu}$ for a rotation by θ_z has non-zero entries $\delta\omega_{12} = -\delta\omega_{21} = \theta_z$, and a boost β_x has $\delta\omega_{10} = -\delta\omega_{01} = \beta_x$.

Adding spacetime translations, generated by $P^\mu = (H, \vec{P})$ with $U^\dagger(\Lambda) P^\mu U(\Lambda) = \Lambda^\mu_\nu P^\nu$, gives the **Poincaré algebra**: the Lorentz algebra plus

$$\begin{aligned} [J_i, H] &= 0, & [J_i, P_j] &= i\hbar \varepsilon_{ijk} P_k, & [K_i, H] &= i\hbar c P_i, & [K_i, P_j] &= \frac{i\hbar}{c} \delta_{ij} H, \\ [H, P_i] &= 0, & [P_i, P_j] &= 0. \end{aligned}$$

Finite transformations are obtained by exponentiation (not very useful in practice): $U(\Lambda) = \exp(\frac{i}{2\hbar} \delta\omega_{\mu\nu} M^{\mu\nu})$ and $T(a) = \exp(-\frac{i}{\hbar} P^\mu a_\mu)$. Fields transform as

$$\begin{aligned} T^\dagger(a) \phi(x) T(a) &= \phi(x - a), & U^\dagger(\Lambda) \phi(x) U(\Lambda) &= \phi(\Lambda^{-1}x), \\ U^\dagger(\Lambda) A^\mu(x) U(\Lambda) &= \Lambda^\mu_\nu A^\nu(\Lambda^{-1}x), \end{aligned}$$

the scalar case following from $\bar{\phi}(\bar{x}) = \phi(x)$ with $\bar{x} = \Lambda x$. Similarly $U^\dagger(\Lambda) \partial^\mu \phi(x) U(\Lambda) = \Lambda^\mu_\rho \bar{\partial}^\rho \phi(\Lambda^{-1}x)$, with the derivative taken with respect to the argument $\bar{x} = \Lambda^{-1}x$.

11.2 Canonical quantisation in the mostly-plus metric

Dirac's idea is to build a quantum theory from a Lagrangian (on which symmetries can be imposed), replacing Poisson brackets $\{r_i, p_j\} = \delta_{ij}$ by commutators $[r_i, p_j] = i\hbar \delta_{ij}$. For a field, $S = \int d^4x \mathcal{L}(\phi, \partial\phi)$, which is Lorentz invariant, and $\Pi := \partial\mathcal{L}/\partial\dot{\phi}$. The free theory

$$S = \int d^4x \left[-\frac{1}{2} (\partial\phi)^2 - \frac{1}{2} m^2 \phi^2 \right]$$

has the expected classical limit, with $[m] = L^{-1}$ and $[\phi] = L^{-1}$ (in $\hbar = c = 1$ units, see below). The equation of motion is the Klein–Gordon equation $-\partial^2 \phi + m^2 \phi = \ddot{\phi} - \nabla^2 \phi + m^2 \phi = 0$, and the Hamiltonian is

$$H = \int d^3x \left[\frac{1}{2} \dot{\phi}^2 + \frac{1}{2} (\nabla\phi)^2 + \frac{1}{2} m^2 \phi^2 \right], \quad [\phi(\vec{x}, t), \Pi(\vec{y}, t)] = i\hbar \delta^3(\vec{x} - \vec{y})$$

(equal-time commutation relation), with $\Pi = \dot{\phi}$. In the Heisenberg picture $\phi(\vec{x}, t) = e^{iHt} \phi(\vec{x}, 0) e^{-iHt}$. Expanding $\phi(\vec{x}, t) = \int d\vec{k} f(\vec{k}, t) e^{i\vec{k} \cdot \vec{x}}$ gives $\ddot{f} + \omega^2 f = 0$, so $f = a(\vec{k}) e^{-i\omega t} + b(\vec{k}) e^{i\omega t}$. With $k \cdot x = -\omega t + \vec{k} \cdot \vec{x}$ and the reality condition $b(-\vec{k}) = a^\dagger(\vec{k})$,

$$\phi(x) = \int d\vec{k} \left[a(\vec{k}) e^{ik \cdot x} + a^\dagger(\vec{k}) e^{-ik \cdot x} \right], \quad \Pi(x) = \int d\vec{k} \left[-i\omega a(\vec{k}) e^{ik \cdot x} + i\omega a^\dagger(\vec{k}) e^{-ik \cdot x} \right].$$

The equal-time commutator is reproduced if $[a(\vec{k}), a(\vec{k}')] = 0$ and

$$[a(\vec{k}), a^\dagger(\vec{k}')] = (2\pi)^3 2\omega \delta^3(\vec{k} - \vec{k}'),$$

and the terms with $[a, a]$ and $[a^\dagger, a^\dagger]$ would otherwise produce a time dependence. The vacuum is defined by $a(\vec{k})|0\rangle = 0$ (this builds the Fock space), one-particle states are $|\vec{k}\rangle := a^\dagger(\vec{k})|0\rangle$, with $\langle \vec{k}' | \vec{k} \rangle = (2\pi)^3 2\omega \delta^3(\vec{k} - \vec{k}')$, and

$$H = \int \widetilde{d^3k} \omega a^\dagger(\vec{k}) a(\vec{k}) + E_0, \quad E_0 = \frac{1}{2} \delta^3(\vec{0}) \int d^3k \omega = \infty.$$

The infinite zero-point energy is not dramatic: experimentally only energy differences matter. Then

$$H |\vec{k}\rangle = \sqrt{\vec{k}^2 + m^2} |\vec{k}\rangle + E_0 |\vec{k}\rangle :$$

the $|\vec{k}\rangle$ are free-particle states.

Remark (Bogoliubov transformations). The vacuum is not unique: with $|\alpha|^2 - |\beta|^2 = 1$, the operators

$$a(\vec{k}) = \alpha b(\vec{k}) + \beta^* b^\dagger(-\vec{k}), \quad a^\dagger(\vec{k}) = \alpha^* b^\dagger(\vec{k}) + \beta b(-\vec{k})$$

satisfy the same algebra, $[b(\vec{k}), b^\dagger(\vec{k}')] = 2\omega(2\pi)^3 \delta^3(\vec{k} - \vec{k}')$, but define a different vacuum $b|0_b\rangle = 0$. In terms of b the Hamiltonian is not diagonal, $H \propto \int \widetilde{d^3k} [b^\dagger b + b b^\dagger + b b + b^\dagger b^\dagger]$.

12 The LSZ Reduction Formula

Consider an interacting theory, for instance

$$\mathcal{L} = \underbrace{-\frac{1}{2}(\partial\phi)^2 - \frac{m^2}{2}\phi^2}_{\mathcal{L}_0} + \underbrace{\frac{g}{3!}\phi^3 + \frac{\lambda}{4!}\phi^4 + \dots}_{\text{interactions: "small" corrections}}.$$

In the free theory we can invert the mode expansion,

$$a(\vec{k}) = \int d^3x e^{-ik\cdot x} [i\dot{\phi} + \omega\phi] = i \int d^3x e^{-ik\cdot x} \overleftrightarrow{\partial}_0 \phi, \quad a^\dagger(\vec{k}) = -i \int d^3x e^{ik\cdot x} \overleftrightarrow{\partial}_0 \phi,$$

with $A \overleftrightarrow{\partial}_0 B := A(\partial_0 B) - (\partial_0 A)B$.

Scattering experiments prepare wave packets that are far apart at $t \rightarrow -\infty$ (the state $|\text{in}\rangle = |\psi(-\infty)\rangle$), let them interact, and detect wave packets far apart at $t \rightarrow +\infty$ ($|\text{out}\rangle$). Define the creation operator of a wave packet,

$$a_1^\dagger := \int d^3k f_1(\vec{k}) a^\dagger(\vec{k}, t), \quad f_1(\vec{k}) \propto \exp[-(\vec{k} - \vec{k}_1)^2/4\sigma^2],$$

where now (interacting theory) $a^\dagger(\vec{k}, t) = -i \int d^3x e^{ik\cdot x} \overleftrightarrow{\partial}_0 \phi$ depends on time, since ϕ no longer satisfies the Klein–Gordon equation. At the end we take $\sigma \rightarrow 0$. The initial and final states are

$$|i\rangle := \lim_{t \rightarrow -\infty} a_1^\dagger(t) a_2^\dagger(t) |0\rangle, \quad |f\rangle := \lim_{t \rightarrow +\infty} a_{1'}^\dagger(t) a_{2'}^\dagger(t) |0\rangle,$$

and we want $\langle f|i\rangle$. Now

$$\begin{aligned} a_1^\dagger(+\infty) - a_1^\dagger(-\infty) &= \int_{-\infty}^{\infty} dt \partial_0 a_1^\dagger(t) = -i \int d^3k f_1(\vec{k}) \int d^4x \partial_0 (e^{ik\cdot x} \overleftrightarrow{\partial}_0 \phi) \\ &= -i \int d^3k f_1(\vec{k}) \int d^4x e^{ik\cdot x} (\partial_0^2 + \omega^2) \phi \\ &= -i \int d^3k f_1(\vec{k}) \int d^4x e^{ik\cdot x} (-\partial^2 + m^2) \phi, \end{aligned}$$

where we used $\omega^2 e^{ik \cdot x} = (\vec{k}^2 + m^2) e^{ik \cdot x}$ and integrated $\vec{k}^2 \rightarrow -\nabla^2$ by parts. Hence

$$\begin{aligned} a_1^\dagger(-\infty) &= a_1^\dagger(+\infty) + i \int d^3k f_1(\vec{k}) \int d^4x e^{ik \cdot x} (-\partial^2 + m^2) \phi(x), \\ a_1(+\infty) &= a_1(-\infty) + i \int d^3k f_1(\vec{k}) \int d^4x e^{-ik \cdot x} (-\partial^2 + m^2) \phi(x). \end{aligned}$$

In a free theory the integrals vanish and $a_1^\dagger(-\infty) = a_1^\dagger(+\infty)$. Now

$$\langle f|i \rangle = \langle 0|a_{2'}(+\infty)a_{1'}(+\infty)a_1^\dagger(-\infty)a_2^\dagger(-\infty)|0 \rangle = \langle 0|T a_{2'}(+\infty)a_{1'}(+\infty)a_1^\dagger(-\infty)a_2^\dagger(-\infty)|0 \rangle,$$

since the operators are already time-ordered (T is the **time-ordering operator**: everything to its right is placed in time order). Substituting the relations above, the time ordering moves all $a(-\infty)$ to the right and all $a^\dagger(+\infty)$ to the left, where they annihilate the vacuum. The only surviving term is the one with no creation or annihilation operators left (and then $f_j(\vec{k}) \rightarrow \delta^3(\vec{k} - \vec{k}_j)$):

$$\begin{aligned} \langle f|i \rangle &= i^{n+n'} \prod_{j=1}^n \int d^4x_j e^{ik_j \cdot x_j} (-\partial_j^2 + m^2) \\ &\quad \times \prod_{j'=1}^{n'} \int d^4x'_{j'} e^{-ik'_{j'} \cdot x'_{j'}} (-\partial_{j'}^2 + m^2) \langle 0|T \phi(x_1) \cdots \phi(x_n) \phi(x'_1) \cdots \phi(x'_{n'})|0 \rangle, \end{aligned}$$

the **LSZ reduction formula**: scattering amplitudes are obtained from time-ordered $(n + n')$ -point correlation functions. This requires $\langle 0|\phi(x)|0 \rangle = 0$ and $\langle \vec{k}|\phi(x)|0 \rangle = e^{-ik \cdot x}$, conditions we will impose on the interacting theory.

13 Path Integrals in Quantum Mechanics

13.1 Time evolution and the Dyson formula

The evolution operator is defined by $|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle$ with $U(t_0, t_0) = \mathbb{K}$. From $i\hbar \partial_t |\psi\rangle = H(t) |\psi\rangle$,

$$i\hbar \partial_t U(t, t_0) = H(t) U(t, t_0), \quad U(t, t_0) = \mathbb{K} + \frac{1}{i\hbar} \int_{t_0}^t dt' H(t') U(t', t_0),$$

with $U(t_2, t_1)U(t_1, t_0) = U(t_2, t_0)$ and $U^{-1}(t, t_0) = U(t_0, t) = U^\dagger(t, t_0)$. Iterating gives the Dyson series,

$$U(t, t_0) = \mathbb{K} + \frac{1}{i\hbar} \int_{t_0}^t dt' H(t') + \left(\frac{1}{i\hbar}\right)^2 \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' H(t') H(t'') + \dots$$

In the n -th term the integration region $t_0 < t_1 < \cdots < t_n < t$ is $1/n!$ of the hypercube (for $n = 2$, half of the square, see Fig. 13.1), and inserting the time-ordering operator one can integrate over the whole hypercube:

$$U(t, t_0) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{1}{i\hbar}\right)^n \int_{t_0}^t dt_n \cdots \int_{t_0}^t dt_1 T\{H(t_n) \cdots H(t_1)\} = T \exp \left[\frac{1}{i\hbar} \int_{t_0}^t dt' H(t') \right],$$

the **Dyson formula** (the same as in Part I, now in the Schrödinger picture).

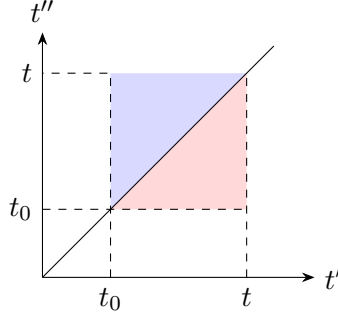


Figure 13.1: The ordered region $t_0 < t'' < t' < t$ is half of the square.

13.2 Derivation of the path integral

Split $U(t_f, t_i) = \prod_{n=1}^N U(t_n, t_{n-1})$ with $\Delta t = (t_f - t_i)/N$ and $U(t_n, t_{n-1}) \simeq \mathbb{1} + \frac{1}{i\hbar} H(t_n) \Delta t$, for $H = \hat{P}^2/2m + V(\hat{X}, t)$. The bases $|p\rangle$ and $|x\rangle$ are not simultaneously diagonalisable ($[X, P] = i\hbar$), but we can insert $\int dp |p\rangle \langle p| = \mathbb{1} = \int dx |x\rangle \langle x|$, with $\langle x|p\rangle = e^{ipx/\hbar}/\sqrt{2\pi\hbar}$:

$$\begin{aligned} U(t_n, t_{n-1}) &= \int dp_n \int dx_n \int dx_{n-1} \left(\mathbb{1} + \frac{\Delta t}{i\hbar} H(p_n, x_n, t_n) \right) |x_n\rangle \langle x_n| p_n \rangle \langle p_n| x_{n-1} \rangle \langle x_{n-1}| \\ &= \int \frac{dp_n}{2\pi\hbar} \int dx_n \int dx_{n-1} e^{-\frac{i\Delta t}{\hbar} [H(p_n, x_n, t_n) - p_n \dot{x}_n]} |x_n\rangle \langle x_{n-1}|, \end{aligned}$$

where $\dot{x}_n := (x_n - x_{n-1})/\Delta t$ is just a symbol. Multiplying the factors (the inner products $\langle x'_n | x_n \rangle = \delta(x'_n - x_n)$ remove half of the position integrals) and taking $N \rightarrow \infty$,

$$U(t_f, t_i) = \int \mathcal{D}p \mathcal{D}x \exp \left[\frac{i}{\hbar} \int_{t_i}^{t_f} dt (p\dot{x} - H) \right] |x_f\rangle \langle x_i|, \quad \mathcal{D}p = \prod_{n=1}^N \frac{dp_n}{2\pi\hbar}, \quad \mathcal{D}x = \prod_{n=0}^N dx_n,$$

the **functional** or **path integral**; the phase $\Delta t \sum_n (p_n \dot{x}_n - H_n) \rightarrow \int dt L$. Completing the square,

$$H_n - p_n \dot{x}_n = \frac{(p_n - m\dot{x}_n)^2}{2m} - \underbrace{\left(\frac{m}{2} \dot{x}_n^2 - V(x_n, t_n) \right)}_{L_n},$$

and the Gaussian momentum integrals give a (x -independent) normalisation $\mathcal{N}$:

$$U(t_f, t_i) = \mathcal{N} \int \mathcal{D}x e^{\frac{i}{\hbar} S[x]} |x(t_f)\rangle \langle x(t_i)|, \quad \langle x_f | U(t_f, t_i) | x_i \rangle = \mathcal{N} \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x e^{\frac{i}{\hbar} \int_{t_i}^{t_f} dt L}.$$

With Heisenberg states $|x, t\rangle := U^\dagger(t, t_0) |x\rangle$ (so that $\langle x', t | x, t \rangle = \delta(x - x')$), the left-hand side is $\langle x_f, t_f | x_i, t_i \rangle$. Similarly one can show that

$$\langle x_f, t_f | T \{ \hat{x}(t_1) \cdots \hat{x}(t_n) \} | x_i, t_i \rangle = \mathcal{N} \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x x(t_1) \cdots x(t_n) e^{\frac{i}{\hbar} \int dt L} :$$

the path integral turns time-ordered products of operators into products of functions.

13.3 The forced harmonic oscillator

Consider $L = L_0 + xJ(t)$ with $L_0 = \frac{m}{2} \dot{x}^2 - \frac{m}{2} \omega_0^2 x^2$ and a forcing $J(t)$ (a “current”). The eigenfunctions of L_0 are $\psi_n(x) = \langle x | n \rangle = A_n H_n(ax) e^{-a^2 x^2/2}$, with $A_n^2 = a/(2^n n! \sqrt{\pi})$, $a = \sqrt{m\omega_0/\hbar}$ and $|n\rangle = (a^\dagger)^n |0\rangle / \sqrt{n!}$; in particular $\psi_0(x) = \sqrt{a} \pi^{-1/4} e^{-a^2 x^2/2}$. We want the

vacuum-to-vacuum amplitude in the presence of J : the probability amplitude of starting in the vacuum and ending in the vacuum after being perturbed,

$$\langle 0|0\rangle_J := \lim_{t_i \rightarrow -\infty, t_f \rightarrow +\infty} \langle 0|U(t_f, t_i)|0\rangle_J =: W_0[J], \quad W_0[0] = 1.$$

(Similarly, $\langle 1|0\rangle_J = \langle 1|U(t_f, t_i)|0\rangle_J$ is the amplitude to end in the first excited state.) Using the path integral,

$$\begin{aligned} W_0[J] &= \mathcal{N} \int \mathcal{D}x e^{\frac{i}{\hbar} \int dt (L_0 + xJ)} \psi_0^*(x(+\infty)) \psi_0(x(-\infty)) \\ &= \mathcal{N}' \int \mathcal{D}x e^{\frac{i}{\hbar} \int dt (L_0 + xJ)} e^{-\frac{a^2}{2}(x^2(+\infty) + x^2(-\infty))}. \end{aligned}$$

Since $f(+\infty) + f(-\infty) = \lim_{\epsilon \rightarrow 0} \epsilon \int dt f(t) e^{-\epsilon|t|}$, the boundary wave functions amount to adding $i\epsilon x^2$ to the Lagrangian:

$$W_0[J] = \mathcal{N}' \int \mathcal{D}x \exp \left[\frac{i}{\hbar} \int dt \left(\frac{m}{2} \dot{x}^2 - \frac{m}{2} (\omega_0^2 - i\epsilon) x^2 + xJ \right) \right].$$

Integrating by parts (the boundary term $\frac{m}{2} [x\dot{x}]_{-\infty}^{+\infty}$ vanishes for fixed endpoints with $\dot{x}(\pm\infty) = 0$),

$$\int dt (L_0 + xJ + i\epsilon x^2) = -\frac{m}{2} \int dt dt' x(t) A(t, t') x(t') + \int dt x(t) J(t) =: -\frac{m}{2} x^\top A x + x^\top J,$$

with the operator $A(t, t') := [\frac{d^2}{dt^2} + \omega_0^2 - i\epsilon] \delta(t - t')$ and the obvious matrix notation in function space ($v \cdot w = \int dt v(t) w(t)$, $(A \cdot B)(t, t'') = \int dt' A(t, t') B(t', t'')$). Completing the square,

$$\frac{m}{2} x^\top A x - x^\top J = \frac{m}{2} y^\top A y - \frac{1}{2m} J^\top A^{-1} J, \quad y := x - \frac{1}{m} A^{-1} J.$$

The remaining integral over y is the same as for $J = 0$ and is fixed by $W_0[0] = 1$, so

$$W_0[J] = \exp \left[\frac{i}{2m\hbar} J^\top A^{-1} J \right] = \exp \left[\frac{i}{2m\hbar} \int dt dt' J(t) \Delta(t - t') J(t') \right].$$

Here $\Delta := A^{-1}$ is the Green's function of the harmonic oscillator: using $\delta(t - t') = \int \frac{d\omega}{2\pi} e^{-i\omega(t-t')}$,

$$A(t, t') = \int \frac{d\omega}{2\pi} [\omega_0^2 - \omega^2 - i\epsilon] e^{-i\omega(t-t')}, \quad \Delta(t - t') = \int \frac{d\omega}{2\pi} \frac{e^{-i\omega(t-t')}}{\omega_0^2 - \omega^2 - i\epsilon},$$

which satisfies $[\frac{d^2}{dt^2} + \omega_0^2 - i\epsilon] \Delta(t - t') = \delta(t - t')$. The $i\epsilon$ makes the inverse well defined (it is the boundary condition). The poles are at $\omega = \pm(\omega_0 - i\epsilon)$: $+\omega_0$ slightly below the real axis and $-\omega_0$ slightly above it. Closing the contour below for $t > t'$ and above for $t < t'$,

$$\Delta(t - t') = \frac{i}{2\omega_0} e^{-i\omega_0(t-t')} \theta(t - t') + \frac{i}{2\omega_0} e^{i\omega_0(t-t')} \theta(t' - t) = \frac{i}{2\omega_0} e^{-i\omega_0|t-t'|},$$

the **Feynman propagator** of the oscillator.

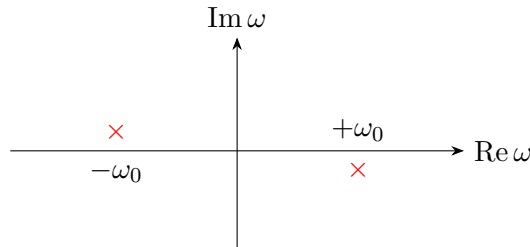


Figure 13.2: Poles of the oscillator propagator with the $i\epsilon$ prescription.

Functional derivatives of $W_0[J]$ give correlation functions; e.g. the amplitude $\langle 1|0\rangle_J$ follows from $\psi_1(x) = H_1(x)\psi_0(x) \propto x\psi_0(x)$ and is proportional to $\delta W_0[J]/\delta J$, while $\frac{\delta}{\delta J(t)} \frac{\delta}{\delta J(t')} W_0[J]|_{J=0} \propto \Delta(t - t')$.

Interactions. For $L = L_0 - V(x) + xJ$,

$$W[J] = \mathcal{N} \int \mathcal{D}x e^{\frac{i}{\hbar} \int dt (L_0 + i\epsilon x^2 + xJ)} e^{-\frac{i}{\hbar} \int dt V(x)} = \sum_{n \geq 0} \frac{1}{n!} \left(-\frac{i}{\hbar} \int dt V \left(\frac{\hbar}{i} \frac{\delta}{\delta J(t)} \right) \right)^n W_0[J],$$

since each $x(t)$ can be generated by $\frac{\hbar}{i} \frac{\delta}{\delta J(t)}$ acting on $e^{\frac{i}{\hbar} \int xJ}$. Therefore

$$W[J] = \exp \left[-\frac{i}{\hbar} \int dt V \left(\frac{\hbar}{i} \frac{\delta}{\delta J(t)} \right) \right] W_0[J],$$

$$W_0[J] = \exp \left[\frac{i}{2m\hbar} \int dt dt' J(t) \Delta(t-t') J(t') \right].$$

14 Path Integrals for Scalar Fields and Feynman Diagrams

14.1 From mechanics to fields

The generalisation to fields is a change of dictionary:

quantum mechanics	field theory
$x(t)$	$\phi(x), x = (t, \vec{x})$
$\mathcal{D}x$	$\mathcal{D}\phi$
$J(t)$	$J(x)$
$L(t)$	$\mathcal{L}(x) = \mathcal{L}_0 + \mathcal{L}_{\text{int}}, \mathcal{L}_{\text{int}} = -V_{\text{int}}$
$\int dt$	$\int d^d x$
$\Delta(t-t')$	$\Delta(x-y)$
W	Z

For $\mathcal{L}_0 = -\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2$,

$$\Delta(x-y) = \int \frac{d^d k}{(2\pi)^d} \frac{e^{ik \cdot (x-y)}}{k^2 + m^2 - i\epsilon}, \quad k^2 = g_{\mu\nu} k^\mu k^\nu,$$

$$Z_0[J] = \exp \left[\frac{i}{2} \int d^d x d^d y J(x) \Delta(x-y) J(y) \right], \quad Z[J] = \exp \left[i \int d^d x \mathcal{L}_{\text{int}} \left(\frac{1}{i} \frac{\delta}{\delta J(x)} \right) \right] Z_0[J].$$

14.2 Renormalised Lagrangian and counterterms

We study the theory (Srednicki's toy model, see Section 15 for why $d = 6$)

$$\mathcal{L} = -\frac{1}{2}(\partial\phi_0)^2 - \frac{m_0^2}{2}\phi_0^2 + \frac{g_0}{3!}\phi_0^3$$

(the **bare Lagrangian**). In an interacting theory the parameters in the Lagrangian are not what is measured: the bare coupling g_0 may be infinite while the observed one is finite (the interactions “dress” the particle, as in Fig. 14.1). Writing the bare quantities in terms of renormalised ones,

$$\phi_0 = Z_\phi^{1/2} \phi, \quad m_0 = Z_\phi^{-1/2} Z_m^{1/2} m, \quad g_0 = Z_\phi^{-3/2} Z_g g,$$

and adding a linear term (needed below),

$$\mathcal{L} = -\frac{1}{2} Z_\phi (\partial\phi)^2 - \frac{Z_m}{2} m^2 \phi^2 + \frac{Z_g}{3!} g \phi^3 + Y \phi = -\frac{1}{2} (\partial\phi)^2 - \frac{m^2}{2} \phi^2 + \frac{g}{3!} \phi^3 + \mathcal{L}_{\text{ct}},$$

with the **counterterms**

$$\mathcal{L}_{\text{ct}} = -\frac{1}{2}(Z_\phi - 1)(\partial\phi)^2 - \frac{1}{2}(Z_m - 1)m^2\phi^2 + \frac{Z_g - 1}{3!}g\phi^3 + Y\phi.$$

They are a toolbox to deal with infinities, characteristic of non-linear theories. (The quantum-mechanical analogue is the anharmonic oscillator $L = \frac{m}{2}\dot{x}^2 - \frac{m\omega_0^2}{2}x^2 + \alpha x^3$.)

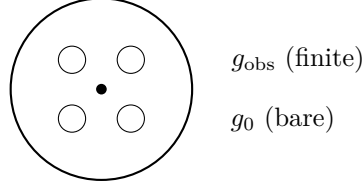


Figure 14.1: A bare charge surrounded by a cloud of virtual pairs: what is observed is the dressed coupling.

14.3 Feynman diagrams from $Z[J]$

Ignoring the counterterms for the moment,

$$\begin{aligned} Z_1[J] &= \exp \left[\frac{ig}{6} \int d^d x \left(\frac{1}{i} \frac{\delta}{\delta J(x)} \right)^3 \right] Z_0[J] \\ &\propto \sum_{V=0}^{\infty} \frac{1}{V!} \left[\frac{ig}{6} \int d^d x \left(\frac{1}{i} \frac{\delta}{\delta J(x)} \right)^3 \right]^V \\ &\quad \times \sum_{P=0}^{\infty} \frac{1}{P!} \left[\frac{i}{2} \int d^d y d^d z J(y) \Delta(y-z) J(z) \right]^P. \end{aligned}$$

Each term is represented by a diagram with the rules

- a source, $i \int d^d x J(x)$: a dot $\bullet$;
- a propagator, $\frac{1}{i} \Delta(x-y)$: a line from x to y ;
- a vertex, $ig \int d^d x$: three lines meeting at a point;
- each diagram is divided by its **symmetry factor**.

For instance,

$$\bullet \text{---} \bullet = \frac{1}{2} \int d^d x d^d y iJ(x) \frac{1}{i} \Delta(x-y) iJ(y), \quad Z_0[J] = e^{\bullet \text{---} \bullet} = \sum_P \frac{1}{P!} (\bullet \text{---} \bullet)^P,$$

where $\frac{1}{2}$ is the symmetry factor. Functional derivatives remove sources:

$$\frac{1}{i} \frac{\delta}{\delta J(x)} \bullet \text{---} \bullet = \int d^d y iJ(y) \frac{1}{i} \Delta(y-x) =: \bullet \text{---} \times_x, \quad \frac{\delta}{\delta J(x)} \frac{\delta}{\delta J(y)} \bullet \text{---} \bullet \propto \Delta(x-y).$$

Products of diagrams count as separate diagrams: two identical disconnected lines form a single diagram with symmetry factor 2, $\frac{1}{2}(\bullet \text{---} \bullet)^2$, three with $3!$, and so on, so $Z_0[J] = 1 + \bullet \text{---} \bullet + \frac{1}{2!}(\bullet \text{---} \bullet)^2 + \frac{1}{3!}(\bullet \text{---} \bullet)^3 + \dots$

First order in g . Acting with three functional derivatives on $e^{\bullet \text{---} \bullet}$,

$$\left(\frac{1}{i} \frac{\delta}{\delta J(x)} \right)^3 e^{\bullet \text{---} \bullet} = \left[3 \Delta_{xx} \bullet \text{---} \times_x + (\bullet \text{---} \times_x)^3 \right] e^{\bullet \text{---} \bullet},$$

with

$$\Delta_{xx} := \frac{1}{i} \Delta(x-x) = \frac{1}{i} \Delta(0) \text{ (a closed loop).}$$

Multiplying by $\frac{ig}{3!} \int d^d x$, the first term is the **tadpole** $\bullet \text{---} \bigcirc$ (symmetry factor 2), which is infinite and will be cancelled with counterterms, and the second is the three-point vertex $\bigwedge$ (symmetry factor 3!):

$$\frac{ig}{3!} \int d^d x \, d^d y_1 \, d^d y_2 \, d^d y_3 \, \Delta(y_1-x) \Delta(y_2-x) \Delta(y_3-x) J(y_1) J(y_2) J(y_3).$$

At second order the same procedure produces vacuum bubbles (two loops joined by a line, the “theta” diagram), propagator corrections (a line with a loop inserted), tree diagrams with four sources, and disconnected products of lower order diagrams.

Connected diagrams. Collecting all terms, every disconnected diagram appears as a product of connected ones with exactly the factors of the exponential series, e.g. $\frac{1}{2} (\bullet \text{---} \bigcirc)^2$, $\frac{1}{3!} (\bigwedge)^3$, $\frac{1}{2} (\bullet \text{---} \bigcirc + \bigwedge)^2$, ... Hence

$$Z[J] = \exp \left[\bullet \text{---} \bullet + \bigwedge + \bullet \text{---} \bigcirc + \dots \right] = e^{iW[J]}, \quad iW[J] = \ln Z[J] = \text{sum of connected diagrams.}$$

In $g\phi^3$ theory all vertices have three legs.

Counterterms and tadpole cancellation. With $\mathcal{L}_{\text{ct}}$, $Z[J] = \exp[i \int \mathcal{L}_{\text{ct}}(\frac{1}{i} \frac{\delta}{\delta J})] Z_1[J]$: the linear term $Y\phi$ is a new one-leg vertex, and the $(Z_\phi - 1)$ and $(Z_m - 1)$ terms are two-leg vertices, which in momentum space give $-i(Ak^2 + Bm^2)$ with $A = Z_\phi - 1$, $B = Z_m - 1$ (drawn as a cross on a line). We choose Y to cancel all diagrams with a single source (tadpoles), so that $\langle 0|\phi|0\rangle = 0$ as required by LSZ. At lowest order

$$iY + \frac{1}{2}(ig)\frac{1}{i}\Delta(0) = 0 \Rightarrow Y = \frac{i}{2}g\Delta(0) + \mathcal{O}(g^3).$$

From now on we can ignore all diagrams containing tadpoles. Loops are where the divergences appear.

Remark (Why the linear term?). For a potential $V(\phi) = \frac{m^2}{2}\phi^2 + \frac{g}{3!}\phi^3 + \frac{\lambda}{4!}\phi^4 + \dots$ the theory is studied near the minimum at $\phi = 0$; if the potential is not bounded below, tunnelling to the true minimum is a non-perturbative problem that perturbation theory does not see. Even near $\phi = 0$, quantum fluctuations shift the potential. In the mechanical analogue $L = \frac{m}{2}\dot{x}^2 - V$ with $V = \frac{\omega_0^2}{2}x^2 + \frac{g}{3!}x^3$ (with $V(0) = V'(0) = 0$),

$$\langle V'(\hat{x}) \rangle = \frac{g}{2} \langle 0|\hat{x}^2|0\rangle \neq 0, \quad \langle V''(\hat{x}) \rangle = \omega_0^2 + g \langle 0|\hat{x}|0\rangle, \quad \langle V'''(\hat{x}) \rangle = g,$$

so the observed potential $V_{\text{obs}} = \langle V \rangle + \langle V' \rangle x + \frac{1}{2} \langle V'' \rangle x^2 + \dots$ differs from the bare one. The counterterm $Y\phi$ removes the induced linear term.

14.4 Scattering amplitude at tree level

By LSZ, $\langle f|i \rangle$ for $2 \rightarrow 2$ requires the four-point function, i.e. four functional derivatives of $Z[J]$; only diagrams with exactly four sources survive. Diagrams in which the two particles propagate freely (two disconnected lines) give the identity, and the scattering matrix is written $S = \mathbb{K} + iT$. At order g^2 the connected diagrams are the s -, t - and u -channel exchanges:

$$\langle f|i \rangle = i(2\pi)^d \delta^d(k_1 + k_2 - k'_1 - k'_2) \mathcal{T},$$

$$\mathcal{T} = g^2 \left[\frac{1}{(k_1 + k_2)^2 + m^2} + \frac{1}{(k_1 - k'_1)^2 + m^2} + \frac{1}{(k_1 - k'_2)^2 + m^2} \right] + \mathcal{O}(g^4),$$

obtained from **amputated** diagrams (the external propagators are removed by the Klein–Gordon operators of LSZ). These are **tree diagrams**; loops appear at order g^4 . With the Mandelstam variable $s = -(k_1 + k_2)^2 = (\omega_1 + \omega_2)^2$ in the centre-of-mass frame, the cross section is

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s |\vec{k}_1|^2} |\mathcal{T}|^2, \quad dt = 2 |\vec{k}_1| |\vec{k}'_1| d\cos\theta.$$

Remark. It is conjectured that $\lambda\phi^4$ theory in four dimensions only exists as an interacting continuum theory for $\lambda = 0$ (“triviality”; the observed coupling would vanish when the whole theory is considered). This is the kind of question addressed by axiomatic (constructive) quantum field theory.

15 Dimensional Analysis and Renormalisability

With $\hbar = c = 1$ there is a single unit, mass: $[M] = 1$, and a quantity with units M^α has $[\cdot] = \alpha$. Since $\hbar = \text{mass} \times \text{velocity} \times \text{length}$ is dimensionless, $[L] = -1$ and $[\partial_\mu] = +1$. (For gravity, $8\pi G_N \sim 1/M_{\text{Pl}}^2$.) The action appears as $e^{iS/\hbar}$, so $[S] = 0$; in d spacetime dimensions, $[d^d x] = -d$, and from $\int d^d x m^2 \phi^2$,

$$[\phi] = \frac{d-2}{2}.$$

For an interaction $\int d^d x \lambda_n \phi^n$, $-d + [\lambda_n] + n \frac{d-2}{2} = 0$, so

$$[\lambda_n] = d - \frac{n(d-2)}{2} = \frac{2d - n(d-2)}{2}.$$

For $g\phi^3$, $[g] = (6-d)/2$; for $\lambda\phi^4$, $[\lambda] = 4-d$. Theories are renormalisable when the coupling has non-negative mass dimension (zero is the borderline case); this is made precise in Section 18. Thus ϕ^3 is renormalisable in $d = 6$ and ϕ^4 in $d = 4$; $g\phi^3$ in $d = 6$ is the simplest renormalisable interacting theory and our toy model.

16 The Exact Propagator and the Self-Energy

16.1 Källén–Lehmann representation

For a free theory, $\Delta(x-y) = i \langle 0 | T \phi(x) \phi(y) | 0 \rangle = \int \frac{d^d k}{(2\pi)^d} e^{ik \cdot (x-y)} \tilde{\Delta}(k^2)$ with $\tilde{\Delta}(k^2) = 1/(k^2 + m^2 - i\epsilon)$. Define the **exact propagator** in the full theory (all interactions switched on) by the same formula,

$$\Delta(x-y) := i \langle 0 | T \phi(x) \phi(y) | 0 \rangle,$$

and impose the LSZ conditions $\langle 0 | \phi | 0 \rangle = 0$ (which fixes Y , removing the tadpoles) and $\langle \vec{k} | \phi(x) | 0 \rangle = e^{-ik \cdot x}$. Insert the identity

$$1 = |0\rangle \langle 0| + \int \widetilde{d\vec{k}} |\vec{k}\rangle \langle \vec{k}| + \sum_n \int \widetilde{d\vec{k}} |\vec{k}, n\rangle \langle \vec{k}, n|,$$

where $|\vec{k}, n\rangle$ are multiparticle states of total momentum $\vec{k}$ (n labels everything else). For $x^0 > y^0$,

$$\langle 0 | \phi(x) \phi(y) | 0 \rangle = \int \widetilde{d\vec{k}} e^{ik \cdot (x-y)} + \sum_n \int \widetilde{d\vec{k}} \langle 0 | \phi(x) | \vec{k}, n \rangle \langle \vec{k}, n | \phi(y) | 0 \rangle.$$

The first term is an integral over the mass hyperboloid $k^0 = \sqrt{\vec{k}^2 + m^2}$. Using $\phi(x) = e^{-iP \cdot x} \phi(0) e^{iP \cdot x}$, $\langle \vec{k}, n | \phi(x) | 0 \rangle = e^{-ik \cdot x} \langle \vec{k}, n | \phi(0) | 0 \rangle$, with $k^0 = \sqrt{\vec{k}^2 + M^2}$ and $M^2 = s$ the invariant mass of the multiparticle state ($s_2 = -(k_1 + k_2)^2$, $s_3 = -(k_1 + k_2 + k_3)^2$, ...). Multiparticle states have $s \geq 4m^2$ (two particles at rest).

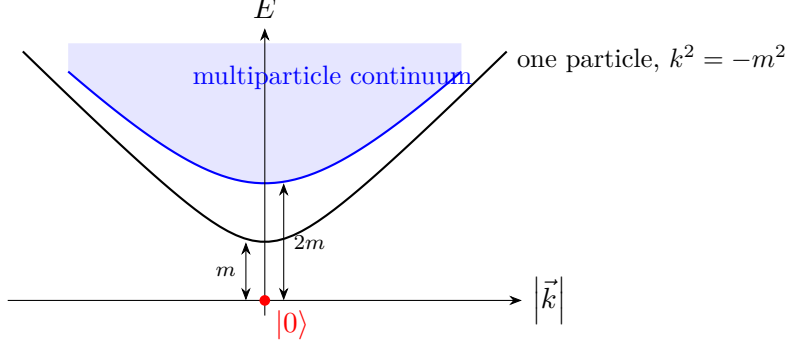


Figure 16.1: Spectrum of the interacting theory: vacuum, one-particle hyperboloid and multiparticle continuum starting at $2m$.

Defining the **spectral density**

$$\rho(s) := \sum_n \left| \langle \vec{k}, n | \phi(0) | 0 \rangle \right|^2 \delta(s - M_n^2) \geq 0,$$

which vanishes for $s < 4m^2$,

$$\langle 0 | \phi(x) \phi(y) | 0 \rangle = \int \widetilde{d}k e^{ik \cdot (x-y)} + \int_{4m^2}^{\infty} ds \rho(s) \int \widetilde{d}k e^{ik \cdot (x-y)} \Big|_{k^0 = \sqrt{\vec{k}^2 + s}},$$

and similarly for $\langle 0 | \phi(y) \phi(x) | 0 \rangle$ with $x - y \rightarrow y - x$. With $-i\Delta = \theta(x^0 - y^0) \langle 0 | \phi(x) \phi(y) | 0 \rangle + \theta(y^0 - x^0) \langle 0 | \phi(y) \phi(x) | 0 \rangle$ and the identity (valid for any mass, the left-hand side being off shell)

$$\int \frac{d^d k}{(2\pi)^d} \frac{e^{ik \cdot (x-y)}}{k^2 + m^2 - i\epsilon} = i\theta(x^0 - y^0) \int \widetilde{d}k e^{ik \cdot (x-y)} + i\theta(y^0 - x^0) \int \widetilde{d}k e^{-ik \cdot (x-y)},$$

with $\widetilde{d}k = d^{d-1}k / ((2\pi)^{d-1} 2k^0)$, we obtain the **Källén–Lehmann representation**

$$\tilde{\Delta}(k^2) = \underbrace{\frac{1}{k^2 + m^2 - i\epsilon}}_{\text{Feynman propagator}} + \underbrace{\int_{4m^2}^{\infty} ds \rho(s) \frac{1}{k^2 + s - i\epsilon}}_{=0 \text{ in a free theory}}.$$

As a function of complex k^2 , $\tilde{\Delta}$ has an isolated pole at $k^2 = -m^2$ and a branch cut from $-4m^2$ to $-\infty$. If $m = 0$, the pole merges with the cut: an **infrared divergence**. Physically, massless particles of very low energy (like photons of very long wavelength) can be indistinguishable from the vacuum.

16.2 Self-energy

Diagrammatically, using $\frac{1}{i}\Delta(x_1 - x_2) = \frac{\delta}{\delta J(x_1)} \frac{\delta}{\delta J(x_2)} iW|_{J=0}$, the exact propagator is the sum of all connected diagrams with two external points: the free line (g^0), the one-loop bubble (g^2), and at order g^4 two bubbles in a chain, the bubble with an internal propagator across it, and so

on. Grouping the **one-particle irreducible** (1PI) diagrams (those that do not fall apart when one internal line is cut) into the **self-energy** $i\Pi(k^2)$,

$$\frac{1}{i}\tilde{\Delta}(k^2) = \frac{1}{i}\tilde{\Delta}_0 + \frac{1}{i}\tilde{\Delta}_0[i\Pi]\frac{1}{i}\tilde{\Delta}_0 + \frac{1}{i}\tilde{\Delta}_0[i\Pi]\frac{1}{i}\tilde{\Delta}_0[i\Pi]\frac{1}{i}\tilde{\Delta}_0 + \dots, \quad \tilde{\Delta}_0 = \frac{1}{k^2 + m^2 - i\epsilon},$$

a geometric series:

$$\tilde{\Delta}(k^2) = \frac{1}{k^2 + m^2 - i\epsilon - \Pi(k^2)}.$$

For this to agree with Källén–Lehmann (a pole at $k^2 = -m^2$ with residue 1) we need $\Pi(-m^2) = 0$ and $\Pi'(-m^2) = 0$. At one loop,

$$i\Pi(k^2) = \frac{1}{2}(ig)^2 \left(\frac{1}{i}\right)^2 \int \frac{d^d l}{(2\pi)^d} \tilde{\Delta}((l+k)^2) \tilde{\Delta}(l^2) - i(Ak^2 + Bm^2) + \mathcal{O}(g^4),$$

the last term coming from the counterterm vertex, with $A = Z_\phi - 1$ and $B = Z_m - 1$.

16.3 Dimensional regularisation

The loop integral diverges in $d = 6$. We compute it in $d = 6 - \varepsilon$ dimensions, replacing $g \rightarrow g\tilde{\mu}^{\varepsilon/2}$ so that g stays dimensionless ($\tilde{\mu}$ is an arbitrary mass scale), with

$$\alpha := \frac{g^2}{(4\pi)^3}, \quad \varepsilon = 6 - d, \quad D := x(1-x)k^2 + m^2.$$

Using Feynman parameters and a Wick rotation (see Section 17),

$$\Pi(k^2) = \frac{1}{2}\alpha \Gamma(-1 + \frac{\varepsilon}{2}) \int_0^1 dx D \left(\frac{4\pi\tilde{\mu}^2}{D} \right)^{\varepsilon/2} - Ak^2 - Bm^2 + \mathcal{O}(\alpha^2).$$

With $A^{\varepsilon/2} = 1 + \frac{\varepsilon}{2} \ln A + \mathcal{O}(\varepsilon^2)$, $\Gamma(-1 + \frac{\varepsilon}{2}) = -\frac{2}{\varepsilon} + \gamma - 1 + \mathcal{O}(\varepsilon)$ and $\int_0^1 dx D = \frac{k^2}{6} + m^2$,

$$\lim_{\varepsilon \rightarrow 0} \Pi(k^2) = -\frac{1}{2}\alpha \left[\left(\frac{2}{\varepsilon} + 1 \right) \left(\frac{k^2}{6} + m^2 \right) + \int_0^1 dx D \ln \left(\frac{4\pi\tilde{\mu}^2}{e^\gamma D} \right) \right] - Ak^2 - Bm^2 + \mathcal{O}(\alpha^2).$$

Defining $\mu := \sqrt{4\pi} e^{-\gamma/2} \tilde{\mu}$,

$$\begin{aligned} \Pi(k^2) = \frac{1}{2}\alpha \int_0^1 dx D \ln \left(\frac{D}{m^2} \right) - \left\{ \frac{1}{6}\alpha \left[\frac{1}{\varepsilon} + \ln \frac{\mu}{m} + \frac{1}{2} \right] + A \right\} k^2 \\ - \left\{ \alpha \left[\frac{1}{\varepsilon} + \ln \frac{\mu}{m} + \frac{1}{2} \right] + B \right\} m^2 + \mathcal{O}(\alpha^2). \end{aligned}$$

The divergences can be absorbed by choosing

$$A = -\frac{1}{6}\alpha \left[\frac{1}{\varepsilon} + \ln \frac{\mu}{m} + \frac{1}{2} + \kappa_A \right] + \mathcal{O}(\alpha^2), \quad B = -\alpha \left[\frac{1}{\varepsilon} + \ln \frac{\mu}{m} + \frac{1}{2} + \kappa_B \right] + \mathcal{O}(\alpha^2),$$

with finite constants κ_A, κ_B , so that $\Pi(k^2) = \frac{1}{2}\alpha \int dx D \ln(D/m^2) + \alpha(\frac{1}{6}\kappa_A k^2 + \kappa_B m^2) + \mathcal{O}(\alpha^2)$. Imposing $\Pi(-m^2) = 0 = \Pi'(-m^2)$ fixes them, and

$$\Pi(k^2) = \frac{1}{2}\alpha \int_0^1 dx D \ln \left(\frac{D}{D_0} \right) - \frac{1}{12}\alpha(k^2 + m^2) + \mathcal{O}(\alpha^2), \quad D_0 := [1 - x(1-x)]m^2.$$

For $k^2 < -4m^2$ the logarithm develops an imaginary part, which is related to the spectral density: from $\tilde{\Delta} = 1/(k^2 + m^2 - i\epsilon - \Pi)$,

$$\rho(s) = \frac{1}{\pi} \frac{\text{Im } \Pi(-s)}{(-s + m^2 - \text{Re } \Pi(-s))^2 + (\text{Im } \Pi(-s))^2}.$$

The conditions $\Pi(-m^2) = 0$ and $\Pi'(-m^2) = 0$ express that the exact propagator has a simple pole at $k^2 = -m^2$ with unit residue.

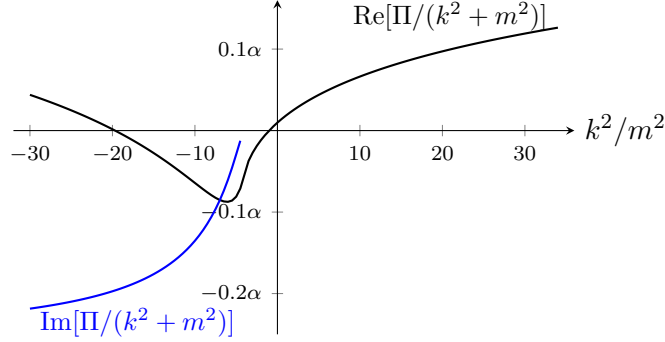


Figure 16.2: One-loop self-energy (on-shell scheme): the imaginary part is non-zero below the threshold $k^2 = -4m^2$, where it is related to $\rho(s)$.

17 Vertex Correction and Loop Integrals

The exact four-point function contains, besides the trivial disconnected parts, tree diagrams, loop corrections on the internal propagator (which we already know how to resum), and corrections to the vertices. Define the **exact three-point vertex** $iV_3(k_1, k_2, k_3)$ (with $k_1 + k_2 + k_3 = 0$) as the sum of 1PI diagrams with three external legs:

$$iV_3(k_1, k_2, k_3) = igZ_g + (ig)^3 \left(\frac{1}{i}\right)^3 \int \frac{d^d l}{(2\pi)^d} \tilde{\Delta}((l - k_1)^2) \tilde{\Delta}((l + k_2)^2) \tilde{\Delta}(l^2) + \mathcal{O}(g^5),$$

with $Z_g = 1 + \mathcal{O}(g^2)$.

Feynman's trick. Combine the denominators,

$$\frac{1}{A_1 A_2 A_3} = \int dF_3 (x_1 A_1 + x_2 A_2 + x_3 A_3)^{-3}, \quad \int dF_3 := 2 \int_0^1 dx_1 dx_2 dx_3 \delta(x_1 + x_2 + x_3 - 1),$$

with $A_1 = (l - k_1)^2 + m^2$, $A_2 = (l + k_2)^2 + m^2$, $A_3 = l^2 + m^2$. Then $x_1 A_1 + x_2 A_2 + x_3 A_3 = q^2 + D$ with

$$q := l - x_1 k_1 + x_2 k_2, \\ D = x_1(1 - x_1)k_1^2 + x_2(1 - x_2)k_2^2 + 2x_1 x_2 k_1 \cdot k_2 + m^2 = x_3 x_1 k_1^2 + x_3 x_2 k_2^2 + x_1 x_2 k_3^2 + m^2,$$

so that

$$\frac{iV_3}{ig} = Z_g + (ig)^2 \left(\frac{1}{i}\right)^3 \int \frac{d^d q}{(2\pi)^d} \int dF_3 (q^2 + D)^{-3}.$$

Wick rotation. The integrand decays at infinity, so the q^0 contour can be deformed avoiding the poles: $q^0 = i\bar{q}_d$, $q^i = \bar{q}_i$, $d^d q = i d^d \bar{q}$ and $q^2 = -(q^0)^2 + (q^1)^2 + \dots = \bar{q}_1^2 + \dots + \bar{q}_d^2$, a Euclidean square. The basic Euclidean integral is

$$\int \frac{d^d \bar{q}}{(2\pi)^d} \frac{(\bar{q}^2)^a}{(\bar{q}^2 + D)^b} = \frac{\Gamma(b - a - \frac{d}{2})\Gamma(a + \frac{d}{2})}{(4\pi)^{d/2}\Gamma(b)\Gamma(\frac{d}{2})} D^{-(b-a-d/2)},$$

convergent if $d + 2a < 2b$. With $g \rightarrow \tilde{\mu}^{\varepsilon/2} g$,

$$\begin{aligned}\frac{V_3(k_1, k_2, k_3)}{g} &= Z_g + \frac{1}{2}\alpha \Gamma\left(\frac{\varepsilon}{2}\right) \int dF_3 \left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\varepsilon/2} + \mathcal{O}(\alpha^2) \\ &= Z_g + \frac{1}{2}\alpha \left[\frac{2}{\varepsilon} + \int dF_3 \ln \left(\frac{4\pi\tilde{\mu}^2}{e^\gamma D} \right) \right] + \mathcal{O}(\alpha^2).\end{aligned}$$

With $\mu^2 = 4\pi e^{-\gamma} \tilde{\mu}^2$ and $Z_g = 1 + C$,

$$\frac{V_3}{g} = 1 + \left\{ \alpha \left[\frac{1}{\varepsilon} + \ln \frac{\mu}{m} \right] + C \right\} - \frac{1}{2}\alpha \int dF_3 \ln \left(\frac{D}{m^2} \right) + \mathcal{O}(\alpha^2),$$

and choosing $C = -\alpha \left[\frac{1}{\varepsilon} + \ln \frac{\mu}{m} + \kappa_C \right] + \mathcal{O}(\alpha^2)$,

$$\frac{V_3(k_1, k_2, k_3)}{g} = 1 - \frac{\alpha}{2} \int dF_3 \ln \left(\frac{D}{m^2} \right) - \kappa_C \alpha + \mathcal{O}(\alpha^2).$$

The condition fixing κ_C is a convention defining g : we take $V_3(0, 0, 0) = g$, the coupling measured at zero momentum (infinite wavelength; experiments fix the momenta as low as possible). Since $D = m^2$ at zero momenta, $\kappa_C = 0$.

Running of the coupling. The coupling depends on the energy scale. If it grows with energy, the theory becomes strongly coupled at high energies and perturbation theory gets worse; if it decreases, perturbation theory improves. QCD is weakly coupled at high energies (asymptotic freedom) and strongly coupled at low energies, which is why quarks are not observed as free particles in our energy range.

Four-point vertex. The box diagram $iV_4(k_1, k_2, k_3, k_4)$ behaves as $\int d^6 l \, l^{-8}$ at large l and is finite:

$$iV_4 = \frac{ig^4}{6(4\pi)^3} \int dF_4 \left(\frac{1}{D_{1234}} + \frac{1}{D_{1324}} + \frac{1}{D_{1423}} \right).$$

Viewed from far away, the box shrinks to a point: an effective four-leg vertex (an *effective theory*). Do new divergences appear at higher orders? The one-loop insertions inside the box (self-energy and vertex corrections) are finite once the counterterms are included, but new divergences do appear in the propagator at higher loops (e.g. the two-loop diagram with a line across the bubble), so $A = Z_\phi - 1$ and $B = Z_m - 1$ acquire terms $\sim g^4/\varepsilon$; the corresponding chains of bubbles and crosses cancel the infinities order by order.

18 Power Counting and the Quantum Action

18.1 Superficial degree of divergence

Summarising: the exact propagator is $\tilde{\Delta}(k^2) = 1/(k^2 + m^2 - i\epsilon - \Pi(k^2))$, with Π the sum of 1PI two-point diagrams, whose divergences are cancelled by A and B ; the exact vertex V_3 is the sum of 1PI three-point diagrams, whose divergences are cancelled by $C = Z_g - 1 = \#g^2 + \#g^4 + \dots$; and V_4 is finite. In general, with the renormalised potential $V(\phi) = \sum_{n \geq 2} \frac{1}{n!} Z_n g_n \phi^n$, define the **superficial degree of divergence** of a diagram

$$\mathcal{D} := dL - 2I,$$

with L the number of loops and I the number of internal propagators: each loop contributes $\int d^d l$ and each propagator $1/l^2$. In $g\phi^3$:

- one-loop vertex (triangle): $\mathcal{D} = d - 6$, so $\mathcal{D} = 0$ in $d = 6$: a logarithmic ultraviolet divergence $\int dl/l$ (the infrared end is not a problem as long as $m \neq 0$);
- one-loop self-energy: $\mathcal{D} = d - 4 = 2$ in $d = 6$: divergences $\propto \infty \cdot k^2 + \infty \cdot m^2$, cancelled by $Ak^2 + Bm^2$;
- two loops ($I = 5$): $\mathcal{D} = 2d - 10$; three loops ($I = 8$): $\mathcal{D} = 3d - 16 = 2$ in $d = 6$: again $\propto k^2, m^2$.

A divergence with $\mathcal{D} = 4$ ($\propto k^4$) would require a new counterterm, i.e. altering the theory with terms like $(\partial^2 \phi)^2$. For $\lambda \phi^4$, the tadpole loop on a propagator has $\mathcal{D} = d - 2$ (it vanishes in dimensional regularisation, see below) and the sunset has $\mathcal{D} = 2d - 6$.

Consider a 1PI diagram with E external legs, L loops, I internal lines and V_n vertices with n legs. It contributes to the coefficient g_E of ϕ^E , so its mass dimension is $[g_E]$. On the other hand, it is $\prod_n g_n^{V_n}$ times an integral of dimension $\mathcal{D}$:

$$[g_E] = dL - 2I + \sum_{n \geq 3} V_n [g_n] \Rightarrow \mathcal{D} = [g_E] - \sum_{n \geq 3} V_n [g_n], \quad [g_n] = d - \frac{1}{2}n(d-2).$$

- If all $[g_n] > 0$: $\mathcal{D}$ decreases with the order, only finitely many diagrams diverge: **super-renormalisable** ($d - \frac{1}{2}n(d-2) > 0$).
- If $[g_n] = 0$: $\mathcal{D} = [g_E]$ is bounded: **renormalisable**.
- If some $[g_n] < 0$: $\mathcal{D}$ grows with the order (e.g. g_5 in $d = 4$), and infinitely many counterterms are needed: **non-renormalisable** ($d - \frac{1}{2}n(d-2) < 0$).

Effective field theories. According to Wilson, all theories look renormalisable at low energies. Higher time derivatives can bring pathologies (causality, ghost fields), so such corrections must be treated as small, e.g.

$$\mathcal{L} = -(\partial\phi)^2 + \frac{\alpha}{\Lambda^2}(\partial^2\phi)^2 + \frac{\alpha'}{\Lambda^4}(\partial\phi)^4 + \dots,$$

with Λ large, so the corrections are small compared with $(\partial\phi)^2$; a theory that resums them is “UV completed”. An example is the Dirac–Born–Infeld Lagrangian $\Lambda^4 \sqrt{1 + (\partial\phi)^2/\Lambda^4}$.

18.2 ϕ^4 theory

For $\mathcal{L} = -\frac{1}{2}(\partial\phi_0)^2 - \frac{m_0^2}{2}\phi_0^2 - \frac{\lambda}{4!}\phi_0^4$ with $\phi_0 = Z_\phi^{1/2}\phi$,

$$\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 - \frac{m^2}{2}\phi^2 - \frac{\lambda Z_\lambda}{4!}\phi^4 - \frac{Z_\phi - 1}{2}(\partial\phi)^2 - \frac{Z_m - 1}{2}m^2\phi^2,$$

with $A = Z_\phi - 1 = a_1\lambda + a_2\lambda^2 + \dots$ and $B = Z_m - 1 = b_1\lambda + b_2\lambda^2 + \dots$. The self-energy has an $\mathcal{O}(\lambda)$ tadpole, $\mathcal{O}(\lambda^2)$ sunset and double-tadpole diagrams, the counterterm $Ak^2 + Bm^2$, and diagrams with counterterm insertions inside loops, e.g. $(i\lambda) \int \frac{d^d l}{(2\pi)^d} (-i)(Al^2 + Bm^2) \frac{1}{(l^2 + m^2)^2}$. In dimensional regularisation $\int d^d l \, l^p = 0$ (scaleless integrals vanish).

18.3 Skeleton expansion and the quantum action

Denote by a double line the exact propagator $\tilde{\Delta}(k^2)$ and by a shaded blob the exact vertex $V(k_1, k_2, k_3)$. The **skeleton expansion** writes any process to all orders using only exact propagators and vertices in diagrams without quantum corrections (skeleton diagrams). For V_4 in ϕ^3 theory, the reducible tree diagram is not part of it (it is already accounted for by exact

propagators and vertices), nor are boxes with self-energy insertions; the box with three-particle-irreducible structure (four cuts are needed to separate it) is. For V_5 , pentagons and boxes with an extra line, and so on. Then one “dresses” the skeleton with exact propagators and vertices. Note that a theory with $V = \frac{g}{3!}\phi^3 + \frac{\lambda}{4!}\phi^4$ is different from one with just $\frac{g}{3!}\phi^3$: in the latter, V_4 is generated by loops.

Can we build a theory whose Feynman rules directly give the exact propagators and vertices? Starting from $S[\phi] = \int d^d x \left[-\frac{1}{2}(\partial\phi)^2 - \frac{m^2}{2}\phi^2 + \frac{g}{3!}\phi^3 \right]$, whose Fourier transform has quadratic kernel $k^2 + m^2 = \tilde{\Delta}_0^{-1}$ and a cubic vertex, define the **quantum action** (effective action)

$$\Gamma[\phi] = -\frac{1}{2} \int \frac{d^d k}{(2\pi)^d} \tilde{\phi}(-k) [k^2 + m^2 - \Pi(k^2)] \tilde{\phi}(k) \\ + \sum_{n=3}^{\infty} \frac{1}{n!} \int \frac{d^d k_1}{(2\pi)^d} \cdots \frac{d^d k_n}{(2\pi)^d} (2\pi)^d \delta^d(k_1 + \cdots + k_n) V_n(k_1, \dots, k_n) \tilde{\phi}(k_1) \cdots \tilde{\phi}(k_n),$$

from which no further quantum corrections are to be computed. Consider

$$Z_{\Gamma}[J] = \int \mathcal{D}\phi \exp \left[\frac{i}{\hbar} \Gamma[\phi] + \frac{i}{\hbar} \int d^d x J\phi \right] = e^{iW_{\Gamma}[J]},$$

where $\hbar$ is here just a small dimensionless parameter (not Planck’s constant). In this theory the propagator is $\hbar \tilde{\Delta}(k^2)$, each vertex carries $\frac{1}{\hbar} V_n$ and each source $\frac{1}{\hbar} J$. A diagram with P propagators, V vertices and E sources scales as $\hbar^{P-V-E} = \hbar^{L-1}$, so

$$W_{\Gamma}[J] = \sum_{L=0}^{\infty} \hbar^{L-1} W_{\Gamma,L}[J], \quad W_{\Gamma,0}[J] = W[J],$$

since the tree diagrams of Γ , built with exact propagators and vertices, reproduce all the connected diagrams of the original theory (true by construction). Taking $\hbar \rightarrow 0$ eliminates the loops: “tree level means classical”. In that limit the path integral is dominated by its stationary point (as $\int dx e^{ikx/\hbar}$),

$$Z_{\Gamma}[J] \simeq \exp \left[\frac{i}{\hbar} \Gamma[\phi_J] + \frac{i}{\hbar} \int d^d x J\phi_J \right] \times (\text{quantum part}), \quad \left. \frac{\delta \Gamma}{\delta \phi} \right|_{\phi_J} = -J,$$

and comparing both sides at order $\hbar^{-1}$,

$$W[J] = \Gamma[\phi_J] + \int d^d x J\phi_J,$$

a **Legendre transformation**. (Compare with the original theory: $Z[J] = \int \mathcal{D}\phi e^{iS[\phi] + i \int J\phi}$, with $\delta[S + \int J\phi] = 0$ the Euler–Lagrange equation; expanding $\phi = \phi_{cl} + \delta\phi$ gives $e^{iS[\phi_{cl}]}$ times a Gaussian, the tree-level diagrams.) Since connected correlation functions are $\langle 0|T\phi(x_1) \cdots \phi(x_n)|0 \rangle_{\text{conn}} = \delta_{J_1} \cdots \delta_{J_n} W|_{J=0}$,

$$\langle 0|\phi(x)|0 \rangle_J = \frac{\delta W}{\delta J(x)} = \int d^d y \frac{\delta \Gamma}{\delta \phi_J(y)} \frac{\delta \phi_J(y)}{\delta J(x)} + \phi_J(x) + \int d^d y J(y) \frac{\delta \phi_J(y)}{\delta J(x)} = \phi_J(x),$$

since $\delta \Gamma / \delta \phi_J = -J$. In a derivative (energy) expansion,

$$\Gamma[\phi] = \int d^d x \left[-\mathcal{U}(\phi) - \frac{1}{2} Z(\phi) (\partial\phi)^2 + \dots \right],$$

where the dots are higher-derivative terms, suppressed at low momenta. This theory is already effective and renormalisable.

19 Infrared Divergences and Observables

Including one loop, the $2 \rightarrow 2$ amplitude in ϕ^3 theory at high energies is

$$\mathcal{T} = \mathcal{T}_0 \left[1 - \frac{11}{12} \alpha \ln \left(\frac{s}{m^2} \right) + \mathcal{O}(m^0, \alpha^2) \right], \quad \mathcal{T}_0 = -g^2 \left(\frac{1}{s} + \frac{1}{t} + \frac{1}{u} \right),$$

where $\mathcal{O}(m^0)$ denotes terms that stay finite in the ultrarelativistic limit. The logarithm is problematic as $m \rightarrow 0$: the massless limit is equivalent to an **infrared divergence**. When $m \rightarrow 0$, the one-particle hyperboloid and the multiparticle continuum merge, so a single particle of momentum $\vec{k}$ becomes indistinguishable from two collinear particles with momenta $\vec{k}/2, \vec{k}/2$.

A detector with finite angular resolution δ cannot distinguish an outgoing particle from one that has split into two nearly collinear particles (an extra propagator and vertex). The processes are distinguishable states in the same Hilbert space, so probabilities (not amplitudes) add:

$$|\mathcal{T}_{\text{obs}}|^2 \widetilde{dk} := |\mathcal{T}|^2 \widetilde{dk} + |\mathcal{T}_{\text{split}}|^2 \frac{\widetilde{dk_1} \widetilde{dk_2}}{2}, \quad \mathcal{T}_{\text{split}} = \left[ig \frac{(-i)}{k^2 + m^2} \right] \mathcal{T},$$

with a symmetry factor $\frac{1}{2}$ and $k = k_1 + k_2$. Using $1 = (2\pi)^{d-1} 2\omega \delta^{d-1}(\vec{k}_1 + \vec{k}_2 - \vec{k}) \widetilde{dk}$,

$$|\mathcal{T}_{\text{obs}}|^2 = |\mathcal{T}|^2 \left[1 + \frac{g^2}{(k^2 + m^2)^2} (2\pi)^{d-1} 2\omega \delta^{d-1}(\vec{k}_1 + \vec{k}_2 - \vec{k}) \frac{\widetilde{dk_1} \widetilde{dk_2}}{2} \right]$$

(summed over the configurations the detector cannot resolve). For $m \rightarrow 0$, $k^2 = (k_1 + k_2)^2 = -4\omega_1\omega_2 \sin^2(\theta/2)$, with θ the angle between $\vec{k}_1$ and $\vec{k}_2$, and $\widetilde{dk_1} \widetilde{dk_2} \sim (\omega_1^{d-3} d\omega_1)(\omega_2^{d-3} d\omega_2) \sin^{d-3} \theta d\theta$, so

$$\frac{\widetilde{dk_1} \widetilde{dk_2}}{(k^2)^2} \sim \frac{d\omega_1}{\omega_1^{5-d}} \frac{d\omega_2}{\omega_2^{5-d}} \frac{d\theta}{\theta^{7-d}},$$

which in $d = 6$ has a logarithmic (collinear) divergence at $\theta \rightarrow 0$. Writing $\vec{k}_1, \vec{k}_2$ at angles β and $\theta - \beta$ from $\vec{k}$, with $|\vec{k}_1| = \frac{\sin(\theta-\beta)}{\sin\theta} |\vec{k}|$, $|\vec{k}_2| = \frac{\sin\beta}{\sin\theta} |\vec{k}|$, and $x = \beta/\theta \in [0, 1]$, the phase-space factor becomes $\frac{\Omega_4}{4(2\pi)^5} \frac{\omega}{\omega_1\omega_2} |\vec{k}_1|^4 d|\vec{k}_1| \sin^3 \beta d\beta$ and, for small angles,

$$k^2 + m^2 = -x(1-x)\vec{k}^2 \left[\theta^2 + \frac{m^2}{\vec{k}^2} f(x) \right], \quad f(x) = \frac{1-x+x^2}{(x-x^2)^2}.$$

Integrating over $0 < \theta < \delta$ (the detector resolution),

$$|\mathcal{T}_{\text{obs}}|^2 = |\mathcal{T}|^2 \left[1 + \frac{g^2 \Omega_4}{4(2\pi)^5} \int_0^1 dx x(1-x) \int_0^\delta d\theta \frac{\theta^3}{[\theta^2 + \frac{m^2}{\vec{k}^2} f(x)]^2} + \dots \right],$$

where the θ integral gives $\frac{1}{2} \ln(\frac{\delta^2 \vec{k}^2}{m^2}) - \frac{1}{2} \ln f(x) - \frac{1}{2}$. Summing the four external legs ($\vec{k}^2 = s/4$ in the centre-of-mass frame),

$$|\mathcal{T}_{\text{obs}}|^2 = |\mathcal{T}|^2 \left[1 + 4 \cdot \frac{1}{12} \alpha \left(\ln \frac{\delta^2 \vec{k}^2}{m^2} + c \right) + \dots \right],$$

with a constant c . Combining with $|\mathcal{T}|^2 = |\mathcal{T}_0|^2 [1 - \frac{11}{6} \alpha \ln(s/m^2) + \dots]$,

$$|\mathcal{T}_{\text{obs}}|^2 = |\mathcal{T}_0|^2 \left[1 - \frac{11}{6} \alpha \ln \frac{s}{m^2} \right] \left[1 + \frac{1}{3} \alpha \ln \frac{\delta^2 s}{m^2} \right] = |\mathcal{T}_0|^2 \left[1 - \alpha \left(\frac{3}{2} \ln \frac{s}{m^2} + \frac{1}{3} \ln \frac{1}{\delta^2} \right) \right].$$

The problem is not completely solved: there is still a $\ln(s/m^2)$, and a dependence on the resolution, which requires incorporating higher-order corrections. The remaining logarithm is dealt with by renormalisation-group methods (next section). (A massless scalar field is in any case problematic: as a remark, the *weak gravity conjecture* states that gravity must be the weakest force.)

20 The Renormalisation Group

20.1 Physical mass and the $\overline{\text{MS}}$ scheme

The physical mass m_{ph} is the location of the pole of the exact propagator; the parameter m in the Lagrangian need not coincide with it. In general,

$$\Pi(k^2) = - \left[A + \frac{\alpha}{6} \left(\frac{1}{\varepsilon} + \frac{1}{2} \right) \right] k^2 - \left[B + \alpha \left(\frac{1}{\varepsilon} + \frac{1}{2} \right) \right] m^2 + \frac{1}{2} \alpha \int_0^1 dx D \ln \left(\frac{D}{\mu^2} \right) + \mathcal{O}(\alpha^2),$$

$$D = x(1-x)k^2 + m^2,$$

and $\Pi'(k^2) = -[A + \frac{\alpha}{6}(\frac{1}{\varepsilon} + \frac{1}{2})] + \frac{\alpha}{2} \int dx x(1-x)[\ln(D/\mu^2) + 1]$. The conditions $\Pi(-m_{\text{ph}}^2) = 0 = \Pi'(-m_{\text{ph}}^2)$ define the on-shell scheme; if $m = 0$ they cannot be imposed as $\Pi(0) = \Pi'(0) = 0$ for any A, B . A different choice is **minimal subtraction** ($\overline{\text{MS}}$), a *renormalisation scheme* in which the counterterms only cancel the $1/\varepsilon$ poles:

$$A = -\frac{1}{6}\alpha\frac{1}{\varepsilon} + \mathcal{O}(\alpha^2), \quad B = -\frac{\alpha}{\varepsilon} + \mathcal{O}(\alpha^2),$$

which gives

$$\Pi_{\overline{\text{MS}}}(k^2) = -\frac{\alpha}{12}(k^2 + 6m^2) + \frac{\alpha}{2} \int_0^1 dx D \ln \left(\frac{D}{\mu^2} \right).$$

The physical mass enters the kinematics, $E = \sqrt{\vec{k}^2 + m_{\text{ph}}^2}$ (a property of the asymptotic states), and by the Källén–Lehmann representation (independent of the theory, a non-perturbative result) $(-\partial_x^2 + m_{\text{ph}}^2)\Delta(x, y) = \delta(x - y)$ near the pole. So m_{ph} is computed from

$$\tilde{\Delta}_{\overline{\text{MS}}}^{-1}(-m_{\text{ph}}^2) = 0, \quad \tilde{\Delta}_{\overline{\text{MS}}}^{-1}(k^2) = k^2 + m^2 - \Pi_{\overline{\text{MS}}}(k^2),$$

whereas $\Pi'(-m_{\text{ph}}^2) \neq 0$ is not a disaster: it only changes the residue (normalisation) R of the pole, so that $\langle 0 | \phi(x) | \vec{k} \rangle = R^{1/2} e^{i\vec{k} \cdot \vec{x}}$ instead of $e^{i\vec{k} \cdot \vec{x}}$. From $-m_{\text{ph}}^2 + m^2 = \Pi_{\overline{\text{MS}}}(-m_{\text{ph}}^2)$ and $m_{\text{ph}}^2 = m^2 + \mathcal{O}(\alpha)$,

$$m_{\text{ph}}^2 = m^2 - \Pi_{\overline{\text{MS}}}(-m^2) + \mathcal{O}(\alpha^2) = m^2 - \frac{\alpha}{2} \left[\frac{m^2}{6} - m^2 + \int_0^1 dx D_0 \ln \left(\frac{D_0}{\mu^2} \right) \right] + \mathcal{O}(\alpha^2),$$

with $D_0 = [1 - x(1-x)]m^2$, i.e.

$$m_{\text{ph}}^2 = m^2 \left[1 + \frac{5}{12} \alpha \left(\ln \frac{\mu^2}{m^2} + c' \right) \right] + \mathcal{O}(\alpha^2), \quad c' = \frac{34 - 3\pi\sqrt{3}}{15} \simeq 1.18.$$

We now have three masses: the bare mass m_0 , the $\overline{\text{MS}}$ mass $m = m(\mu)$ and the physical mass m_{ph} (analogous to the bare frequency, the parameter, and the observed frequency of an anharmonic oscillator). Neither the bare nor the physical mass can depend on the arbitrary scale μ :

$$\frac{dm_0^2}{d\mu} = 0, \quad \frac{dm_{\text{ph}}^2}{d\mu} = 0.$$

From $\ln m_{\text{ph}} = \ln m + \frac{5}{12} \alpha \ln \frac{\mu}{m} + \dots$,

$$0 = \frac{d \ln m_{\text{ph}}}{d \ln \mu} = \frac{1}{m} \frac{dm}{d \ln \mu} + \frac{5}{12} \alpha + \mathcal{O}(\alpha^2) \Rightarrow \frac{dm}{d \ln \mu} = -\frac{5}{12} \alpha m,$$

one of the **renormalisation group equations**. The coupling also runs, $g_0 \rightarrow gZ_g$ with $g = g(\mu)$, so $\alpha = \alpha(\mu)$; this does not affect the previous equation, since $d\alpha/d \ln \mu = \mathcal{O}(\alpha^2)$.

Residue. Define

$$R^{-1} := \frac{d}{dk^2} \left[\tilde{\Delta}_{\overline{\text{MS}}}^{-1}(k^2) \right]_{k^2 = -m_{\text{ph}}^2} = 1 - \Pi'_{\overline{\text{MS}}}(-m_{\text{ph}}^2) = 1 - \Pi'_{\overline{\text{MS}}}(-m^2) + \mathcal{O}(\alpha^2),$$

$$R^{-1} = 1 + \frac{1}{12} \alpha \left[\ln \frac{\mu^2}{m^2} + c'' \right] + \mathcal{O}(\alpha^2), \quad c'' = \frac{17 - 3\pi\sqrt{3}}{3} \simeq 0.23.$$

Similarly, in $\overline{\text{MS}}$, $V_{3,\overline{\text{MS}}}(k_1, k_2, k_3) = g[1 - \frac{\alpha}{2} \int dF_3 \ln(D/\mu^2) + \mathcal{O}(\alpha^2)]$, which is not yet observable, and

$$\mathcal{T} = \mathcal{T}_0 \left[1 - \frac{11}{12} \alpha \left(\ln \frac{s}{\mu^2} + \mathcal{O}(m^0) \right) + \mathcal{O}(\alpha^2) \right].$$

The LSZ formula requires a field normalised as $\langle 0 | \phi_{\text{LSZ}}(x) | \vec{k} \rangle = e^{i\vec{k} \cdot \vec{x}}$, so $\phi(x) = R^{1/2} \phi_{\text{LSZ}}(x)$, with $\Delta_{\overline{\text{MS}}}(x, y) = R \Delta_{\text{LSZ}}(x, y)$ and $\langle 0 | \phi \cdots \phi | 0 \rangle = R^{n/2} \langle 0 | \phi_{\text{LSZ}} \cdots \phi_{\text{LSZ}} | 0 \rangle$ for n fields. Amputating the four external propagators (each carrying a factor R) and correcting the normalisation,

$$\mathcal{T} = R^4 R^{-2} \mathcal{T}_0[\dots] = R^2 \mathcal{T}_0 \left[1 - \frac{11}{12} \alpha \left(\ln \frac{s}{\mu^2} + \mathcal{O}(m^0) \right) + \mathcal{O}(\alpha^2) \right],$$

$$\mathcal{T} = \mathcal{T}_0 \left[1 - \alpha \left(\frac{11}{12} \ln \frac{s}{\mu^2} + \frac{1}{6} \ln \frac{\mu^2}{m^2} + \mathcal{O}(m^0) \right) + \mathcal{O}(\alpha^2) \right].$$

20.2 Running coupling

Including the detector resolution as before,

$$|\mathcal{T}|_{\text{obs}}^2 = |\mathcal{T}|^2 \left[1 + \frac{1}{3} \alpha \ln \frac{\delta^2 s}{m^2} \right] = |\mathcal{T}_0|^2 \left[1 - \alpha(\mu) \left(\frac{3}{2} \ln \frac{s}{\mu^2} + \frac{1}{3} \ln \frac{1}{\delta^2} \right) \right].$$

The observable cannot depend on μ . Since $|\mathcal{T}_0|^2 \propto g^4 \propto \alpha^2$,

$$0 = \frac{d \ln |\mathcal{T}|_{\text{obs}}^2}{d \ln \mu} = \frac{2}{\alpha} \frac{d\alpha}{d \ln \mu} + 3\alpha \Rightarrow \beta(\alpha) := \frac{d\alpha}{d \ln \mu} = -\frac{3}{2} \alpha^2,$$

the **beta function**, with solution

$$\alpha(\mu_2) = \frac{\alpha(\mu_1)}{1 + \frac{3}{2} \alpha(\mu_1) \ln(\mu_2/\mu_1)}.$$

We can take $\mu^2 = s$, which looks dirty but eliminates the large logarithm:

$$\alpha(E_2) = \frac{\alpha(E_1)}{1 + \frac{3}{2} \alpha(E_1) \ln(E_2/E_1)}.$$

As E grows, $g(E)$ decreases: ϕ^3 theory in six dimensions is **asymptotically free**. Extrapolating the running of the couplings of the Standard Model (electromagnetic, weak, strong), α_i^{-1} versus $\ln E$, they do not quite intersect at a single point; a new theory that unifies them would be a *grand unified theory* (GUT).

20.3 Renormalisation group equations from the counterterms

Write the bare Lagrangian with $b = 0$,

$$\mathcal{L} = -\frac{1}{2}(\partial\phi_0)^2 - \frac{1}{2}m_0^2\phi_0^2 + \frac{1}{3!}g_0\phi_0^3 + Y_0\phi_0 = -\frac{1}{2}Z_\phi(\partial\phi)^2 - \frac{1}{2}Z_m m^2\phi^2 + \frac{1}{3!}Z_g\tilde{\mu}^{\varepsilon/2}g\phi^3 + Y\phi,$$

$$\phi_0 = Z_\phi^{1/2}\phi, \quad m_0 = Z_\phi^{-1/2}Z_m^{1/2}m, \quad g_0 = Z_\phi^{-3/2}Z_gg\tilde{\mu}^{\varepsilon/2}, \quad Y_0 = Z_\phi^{-1/2}Y,$$

with $\varepsilon = 6 - d$, $\alpha = g^2/(4\pi)^3$ and $\mu^2 = 4\pi e^{-\gamma}\tilde{\mu}^2$. In $\overline{\text{MS}}$ the Z 's only contain poles,

$$Z_\phi = 1 + \sum_{n=1}^{\infty} \frac{a_n(\alpha)}{\varepsilon^n}, \quad Z_m = 1 + \sum_{n=1}^{\infty} \frac{b_n(\alpha)}{\varepsilon^n}, \quad Z_g = 1 + \sum_{n=1}^{\infty} \frac{c_n(\alpha)}{\varepsilon^n},$$

with $a_1 = -\frac{\alpha}{6} + \mathcal{O}(\alpha^2)$, $b_1 = -\alpha + \mathcal{O}(\alpha^2)$ and $c_1 = -\alpha + \mathcal{O}(\alpha^2)$. Observables are functions of α and μ with $d|\mathcal{T}|_{\text{obs}}/d\mu = 0$; the strategy here is instead to use that bare quantities do not depend on μ . The logic is

$$\mathcal{O}_b \left(d\mathcal{O}_b/d\mu = 0 \right) \longrightarrow \mathcal{O}_{\text{ren}}(\mu) \longrightarrow \mathcal{O}_{\text{obs}} \left(d\mathcal{O}_{\text{obs}}/d\mu = 0 \right).$$

Beta function. With $\alpha_0 := g_0^2/(4\pi)^3 = Z_g^2 Z_\phi^{-3} \tilde{\mu}^\varepsilon \alpha$ and $d \ln \alpha_0 / d \ln \mu = 0$, define

$$G(\alpha, \varepsilon) := \ln(Z_g^2 Z_\phi^{-3}) = \ln \left[\left(1 + \sum_n \frac{a_n}{\varepsilon^n} \right)^{-3} \left(1 + \sum_n \frac{c_n}{\varepsilon^n} \right)^2 \right] = \sum_{n=1}^{\infty} \frac{G_n(\alpha)}{\varepsilon^n},$$

$$G_1 = 2c_1 - 3a_1 = -\frac{3}{2}\alpha + \mathcal{O}(\alpha^2).$$

Then $\ln \alpha_0 = G(\alpha, \varepsilon) + \ln \alpha + \varepsilon \ln \tilde{\mu}$, and differentiating with respect to $\ln \mu$ and multiplying by α ,

$$0 = \frac{d\alpha}{d \ln \mu} \left[1 + \frac{\alpha G'_1(\alpha)}{\varepsilon} + \frac{\alpha G'_2(\alpha)}{\varepsilon^2} + \dots \right] + \alpha \varepsilon, \quad (')' := \frac{d}{d\alpha}.$$

Since $d\alpha/d \ln \mu$ must be finite as $\varepsilon \rightarrow 0$, it has the form $d\alpha/d \ln \mu = -\alpha \varepsilon + \beta(\alpha)$, and the higher poles cancel “telescopically”:

$$0 = \underbrace{\beta(\alpha) - \alpha^2 G'_1(\alpha)}_{=0} + \frac{1}{\varepsilon} \underbrace{(-\alpha^2 G'_2(\alpha) + \beta(\alpha) \alpha G'_1(\alpha))}_{=0} + \dots$$

$$\Rightarrow \beta(\alpha) = \alpha^2 G'_1(\alpha) = -\frac{3}{2}\alpha^2 + \mathcal{O}(\alpha^3),$$

in agreement with the result obtained from the observable.

Anomalous dimension of the mass. Similarly, with

$$M(\alpha, \varepsilon) := \ln(Z_m^{1/2} Z_\phi^{-1/2}) = \sum_{n=1}^{\infty} \frac{M_n(\alpha)}{\varepsilon^n}, \quad M_1 = \frac{1}{2}b_1 - \frac{1}{2}a_1 = -\frac{5}{12}\alpha + \mathcal{O}(\alpha^2),$$

$\ln m_0 = M(\alpha, \varepsilon) + \ln m$, and

$$0 = \frac{d \ln m_0}{d \ln \mu} = \frac{\partial M}{\partial \alpha} \frac{d\alpha}{d \ln \mu} + \frac{1}{m} \frac{dm}{d \ln \mu}$$

$$\Rightarrow \frac{1}{m} \frac{dm}{d \ln \mu} = (\varepsilon \alpha - \beta(\alpha)) \left(\frac{M'_1}{\varepsilon} + \frac{M'_2}{\varepsilon^2} + \dots \right) = \alpha M'_1 + \frac{1}{\varepsilon} \underbrace{(\alpha M'_2 - \beta M'_1)}_0 + \dots$$

The **anomalous dimension** of the mass is

$$\gamma_m(\alpha) := \frac{1}{m} \frac{dm}{d \ln \mu} = \alpha M'_1(\alpha) = -\frac{5}{12}\alpha + \mathcal{O}(\alpha^2),$$

again in agreement with the previous result.

Callan–Symanzik equation. The bare propagator $\tilde{\Delta}_0(k^2) = i \int d^6x e^{-ik \cdot x} \langle 0 | T \phi_0(x) \phi_0(0) | 0 \rangle$ is related to the renormalised one, $\tilde{\Delta}(k^2) = i \int d^6x e^{-ik \cdot x} \langle 0 | T \phi(x) \phi(0) | 0 \rangle$, by $\tilde{\Delta}_0(k^2) = Z_\phi \tilde{\Delta}(k^2)$. Since the bare propagator does not depend on μ ,

$$0 = \frac{d \ln \tilde{\Delta}_0}{d \ln \mu} = \frac{d \ln Z_\phi}{d \ln \mu} + \frac{1}{\tilde{\Delta}} \left[\frac{\partial}{\partial \ln \mu} + \frac{d\alpha}{d \ln \mu} \frac{\partial}{\partial \alpha} + \frac{dm}{d \ln \mu} \frac{\partial}{\partial m} \right] \tilde{\Delta}(k^2).$$

With $\ln Z_\phi = \frac{a_1}{\varepsilon} + \frac{1}{\varepsilon^2} (a_2 - \frac{1}{2} a_1^2) + \dots$,

$$\frac{\partial \ln Z_\phi}{\partial \alpha} \frac{d\alpha}{d \ln \mu} = \left(\frac{a'_1}{\varepsilon} + \dots \right) (-\varepsilon \alpha + \beta(\alpha)) = -\alpha a'_1 + \mathcal{O}(1/\varepsilon),$$

which gives the **Callan–Symanzik equation**

$$\left(\frac{\partial}{\partial \ln \mu} + \beta(\alpha) \frac{\partial}{\partial \alpha} + \gamma_m(\alpha) m \frac{\partial}{\partial m} + 2\gamma_\phi(\alpha) \right) \tilde{\Delta}(k^2) = 0,$$

$$\gamma_\phi(\alpha) := \frac{1}{2} \frac{d \ln Z_\phi}{d \ln \mu} = -\frac{\alpha a'_1}{2} = \frac{1}{12} \alpha + \mathcal{O}(\alpha^2),$$

with γ_ϕ the anomalous dimension of the field. At a fixed point, $\beta(\alpha_*) = 0$, and for $m = 0$, the solution is

$$\tilde{\Delta}(k^2) \sim \left(\frac{\mu^2}{k^2} \right)^{-\gamma_\phi(\alpha_*)} \frac{C(\alpha_*)}{k^2},$$

a power law with an anomalous exponent. (See also Peskin & Schroeder.)